LANDSCAPE ARCHITECTURE

PORTFOLIO

TIANYE WU

Park Equity & Access / Urban Heat & Climate Adaptation / Ecological Remediation / Urban Renewal / Visual Communication

Master of Landscape Architecture, Cornell University

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ABOUT

Landscape architecture student working between spatial research and design, with a background in visual communication.

EDUCATION

Cornell University · Master of Landscape Architecture, 2025–2027

Shanghai Jiao Tong University · B.Des. Visual Communication Design, 2021–2025

RESEARCH

First author, “Neighborhood Socio-Racial Composition and the Spatial Reach of Park Visitation in New York City,” *Land* (2026)

Master’s thesis (ongoing): NYC park renovations, visitor origins, and local benefit capture

EXPERIENCE

Design Intern, Hangzhou Urban Transport Planning and Design Institute, 2024

SOFTWARE

Python · ArcGIS · QGIS · Rhino · Grasshopper · AutoCAD · SketchUp · Illustrator · InDesign · Photoshop · After Effects · Enscape · D5 Render

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RESEARCH

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Neighborhood Socio-Racial Composition and the Spatial Reach of Park Visitation in New York City

Tianye Wu, Jie Kong, Xiwei Shen. *Land* 2026, 15(10), 1818.
doi.org/10.3390/land15101818

QUESTION

Static park-access measures show who lives near a park, not how far people actually travel to use one. How far does observed park visitation reach from home, and how does that reach differ with neighborhood socio-racial composition?

DATA & METHODS

2023 Advan Weekly Patterns (52 weeks), ACS, NYC Parks and ParkServe inventories, EPA Smart Location Database. The **Local Park-Use Ratio (LPUR)** is the share of a block group's dwell-qualified park visits that stay within 2 km. Global and local Moran statistics and partial spatial Durbin models, with radius-sensitivity and robustness checks.

KEY FINDINGS

- LPUR is strongly clustered (Moran's I = 0.718).
- An IQR increase in Non-White share is associated with lower LPUR (direct −0.022; total −0.065) and longer trips to parks (total +0.75 km).
- Park quality and local park supply are positively associated with LPUR.

Findings describe neighborhood composition, not individual behavior, and cannot separate preference from constraint.

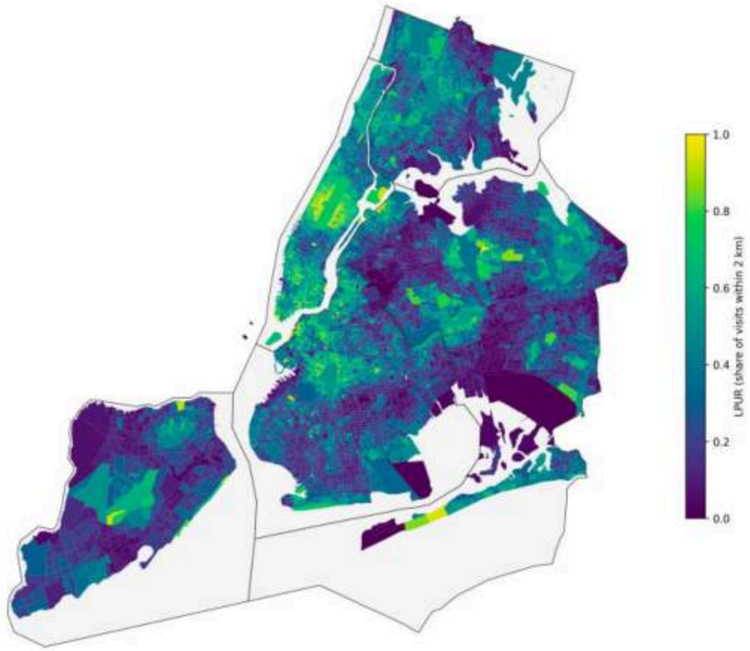


Fig. 3 LPUR by block group: share of park visits within 2 km of home.

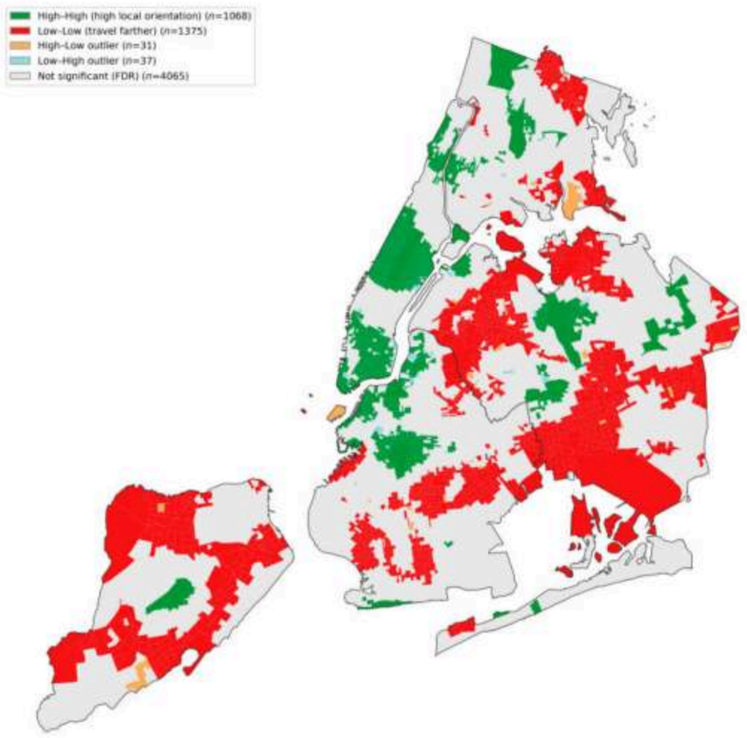


Fig. 4 Local Moran clusters (FDR q < 0.05). Green: high local use; red: residents travel farther.

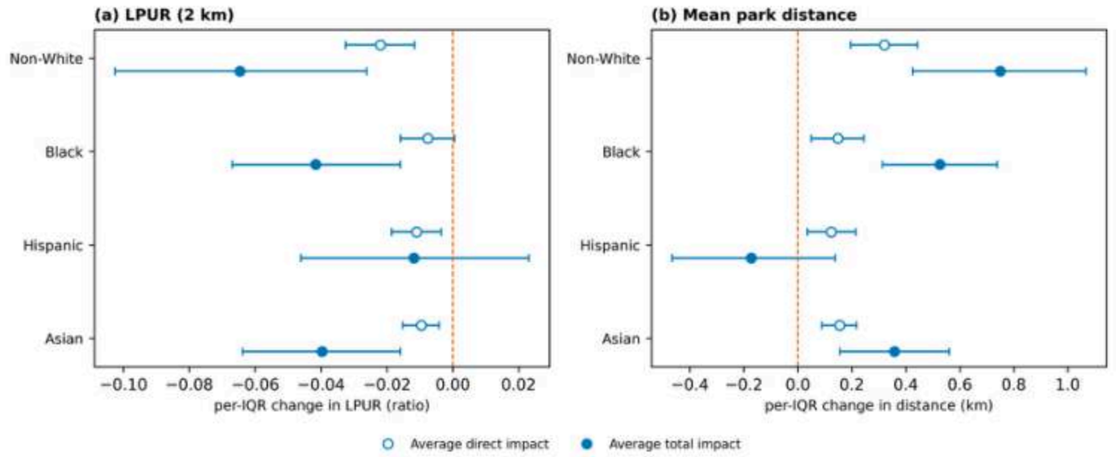


Fig. 6 Adjusted associations of racial composition with LPUR and mean park distance (per-IQR change, 95% CI). Hollow: direct; filled: total impact.

Next step · Master's thesis (in progress): how park renovations in NYC change visitor origins, and who captures the benefits of park investment.

The Living Armature

An urban cooling network for Nodine Hill, Yonkers, NY

Nodine Hill is a dense, low-canopy neighborhood where streets carry both daily pedestrian life and stormwater. We mapped where **direct sun, runoff, and movement coincide**, then built a continuous green "armature" of **cooling corridors** and **street-tree typologies matched to sun exposure**.

Group project with Eliza Li and Kexin Cao (pp. 04–09).
Design #04 Terrace Garden (pp. 10–13) is my individual work.
Rhino + Ladybug · Shademap · GIS · Adobe Suite

≈85% → ≈70%

study area under **high direct sun** (11.2–14 h, June 21)

38 → 210

trees in the study area

49 → 36 ft

average **street-tree spacing**

+2.78%

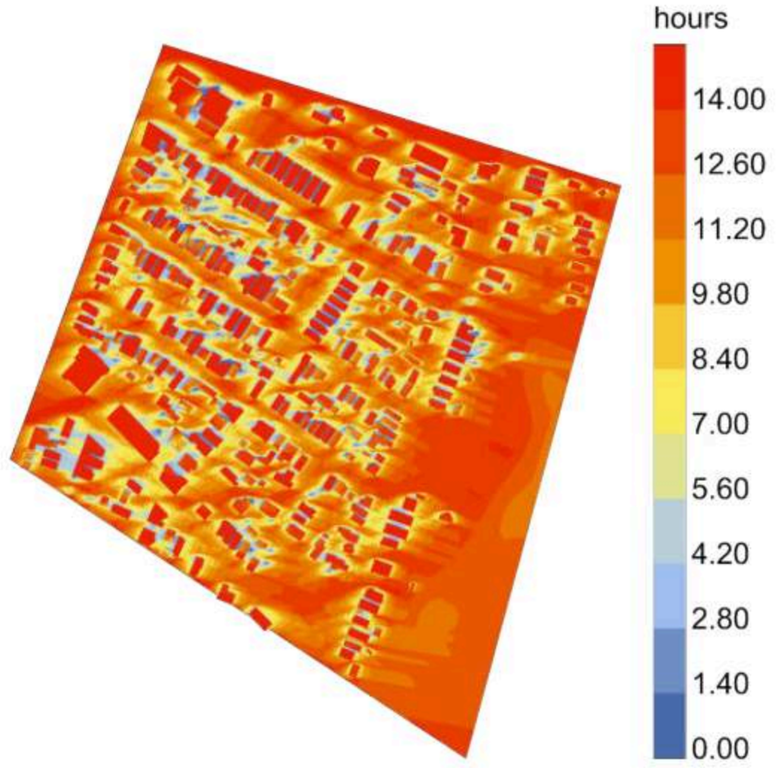
on-site **stormwater** capture / delay



Where the heat is



Hot spots, shaded patches, and flooding spots. Base: Shademap.app; hot spots and patches mapped by the team.



Direct sun hours, existing. Rhino + Ladybug, June 21.

READING THE MAPS

1 Sun everywhere. Most of the study area receives 11–14 hours of direct sun on June 21; only narrow strips beside buildings stay under four hours.

2 Hot streets. Hot spots follow the main streets, where canopy is sparse and pavement is continuous.

3 Heat and water overlap. The largest hot area, the open ground around Fleming Park, also holds the main flooding spots, so heat and runoff can be treated together.

4 No place to cool down. Existing shaded patches are small and disconnected along walking routes.

A network, not isolated parks



Focus areas where cooling (C), ecological (E), and stormwater (S) needs overlap.

Instead of one large cooling park, the plan **threads shade through the streets people already use**. Focus areas become **cooling nodes** at the water tower, a corner lot, the community-center lot, and the terrace garden.

49.4 → 35.8 ft

average spacing between street trees, existing → proposed

3 + 4

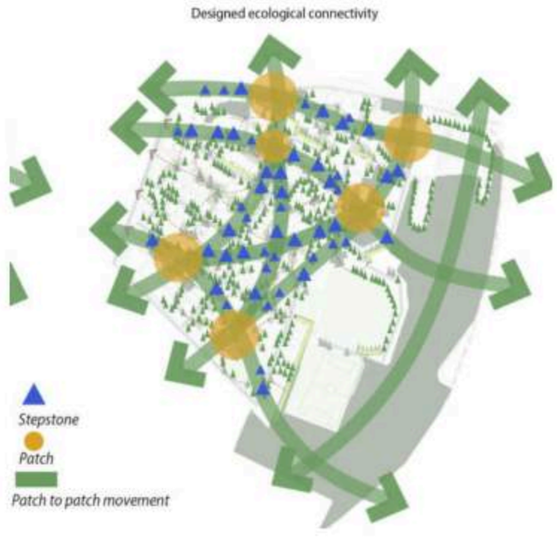
cooling corridors + shaded bus stops

TWO NETWORKS, ONE ARMATURE

Cooling corridors follow where people walk; **habitat corridors** follow how birds move between green patches. Where they overlap, **one line of street trees does both jobs**.



Cooling corridors along walking and school routes, with shaded bus stops.



Habitat corridors link green patches through stepping stones. Denser planting lets the two networks connect.

Reading each street before planting it



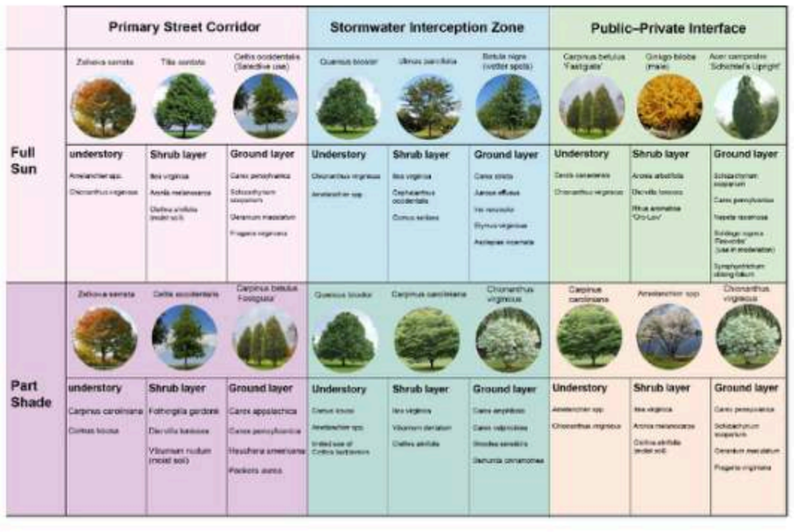
3 · Planting plan. Every proposed tree is placed by typology; colors match the six types on the next page.



1 · Constraints. Each segment is read as a primary street corridor, stormwater interception zone, or public–private interface, together with utility poles and power lines.

2 · SIX TYPOLOGIES

Crossing the three settings with the sun-hour results (**full sun** or **part shade**) gives six typologies, each with its own species palette. Under power lines only **small trees (under 20–25 ft)** fit.



Planting matched to sun exposure

The six typologies in detail. Dot colors match the planting plan. In each pair, existing is on the left and proposed on the right.

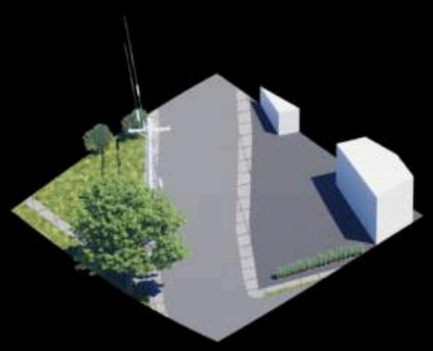
Primary street corridor

Exposed corridors with fragmented planting.
Build a continuous canopy first, then add a middle layer to soften the street edge.

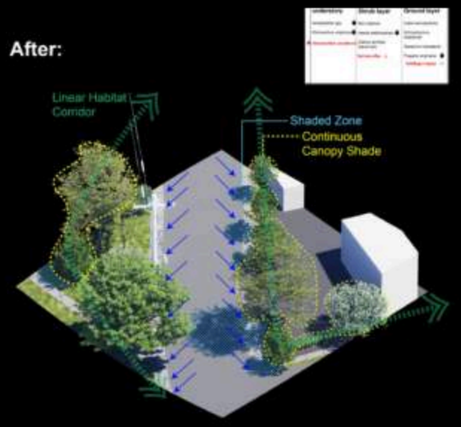
Full sun Part shade

FULL SUN

Before:

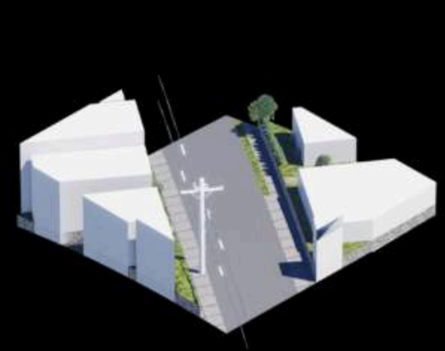


After:

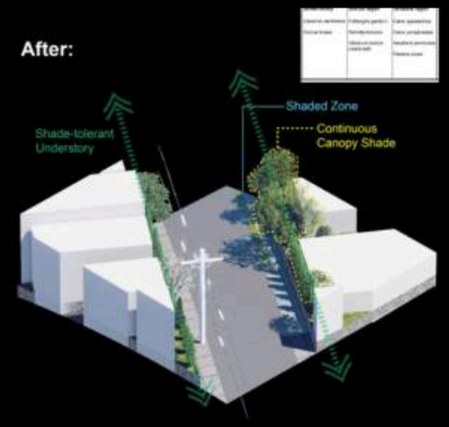


PART SHADE

Before:



After:

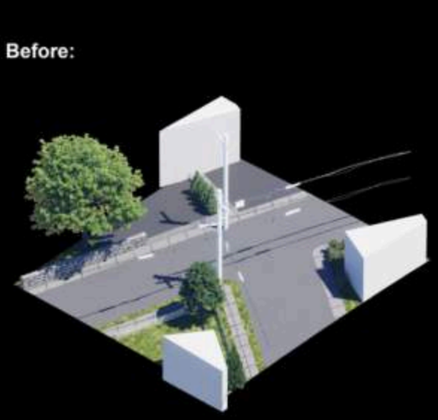


Stormwater interception zone

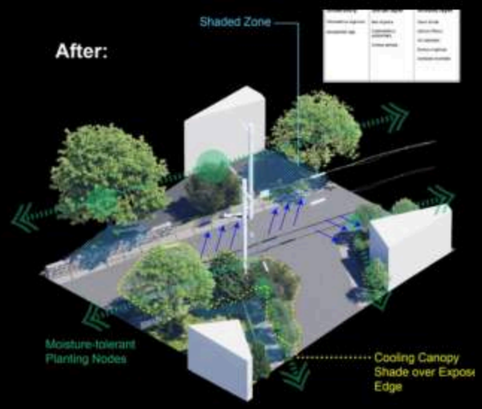
Runoff crosses hard surfaces straight into drains.
Runoff-tolerant trees and layered planting slow and spread the water.

Full sun Part shade

Before:



After:



Before:



After:

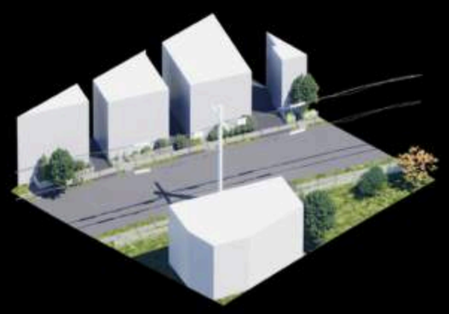


Public–private interface

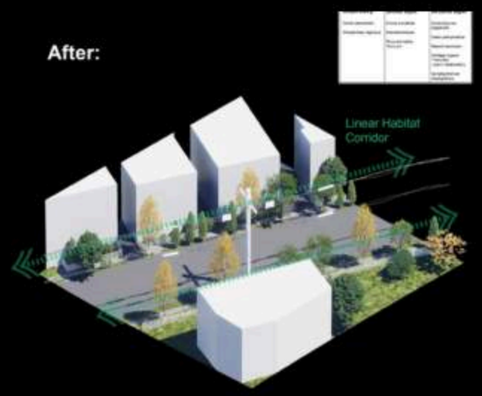
An abrupt edge between sidewalk and front yards.
Controlled tree forms make a continuous edge; middle-layer planting softens the transition.

Full sun Part shade

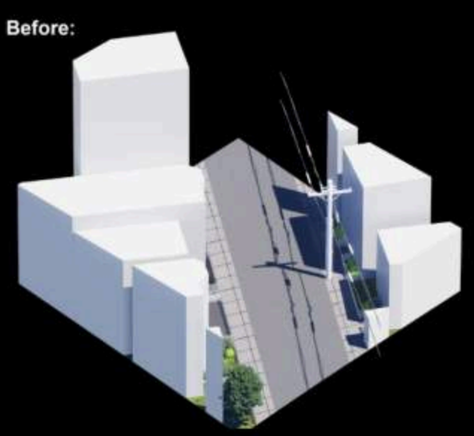
Before:



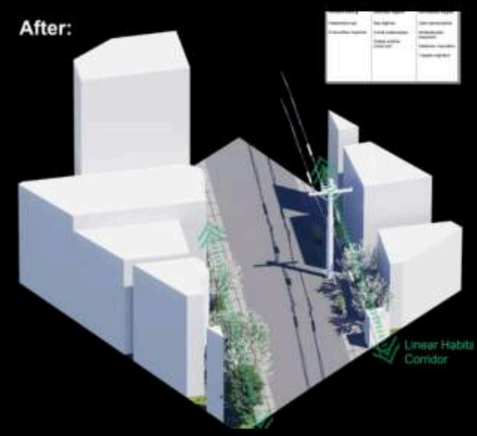
After:



Before:



After:



Measuring the cooling effect

Re-running the June 21 sun-hour simulation with the proposed canopy, **high-exposure area falls by about 15 percentage points** and **low-exposure area roughly triples**.



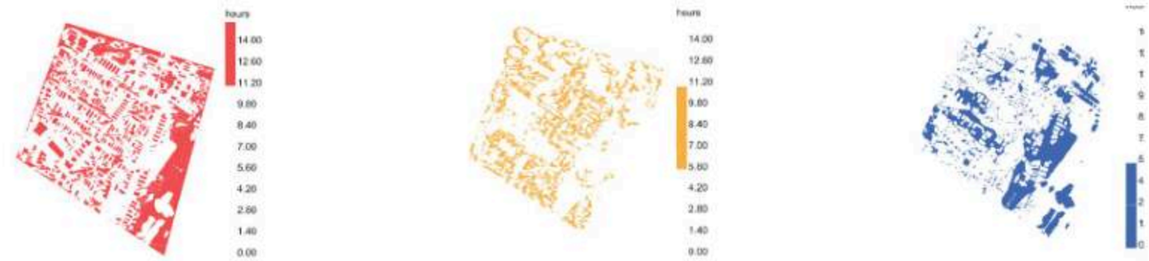
Direct sun hours, June 21. Left: existing. Right: with the proposed tree canopy. Rhino + Ladybug.

SHARE OF STUDY AREA BY DIRECT-SUN EXPOSURE



Exposure ranges are area take-offs from the sun-hour maps, assisted by generative AI (ChatGPT) as noted on the original studio boards.

PROPOSED CONDITION BY EXPOSURE CLASS



38 → 210

trees in the study area

≈29% → ≈61%

native share of the canopy

+2.78%

on-site **stormwater** capture / delay

A terraced garden as integrated cooling infrastructure

The terrace garden is designed for **integrated management**. Its **primary system is heat**: one continuous canopy shades a pedestrian garden, a terraced main passage, and a fitness terrace. **Secondary systems** handle stormwater, through rain gardens along the street edge, and habitat, through layered native planting.

Before:







1. Shaded Pedestrian Garden

This view shows the pedestrian area within the terraced garden. The wooden deck surface creates a warm and accessible walking experience, while flowering plants and layered planting soften the edge of the path. Tree canopy and dense vegetation provide shade, turning the walkway into a comfortable park setting rather than only a circulation route.



2. Terraced Main Passage

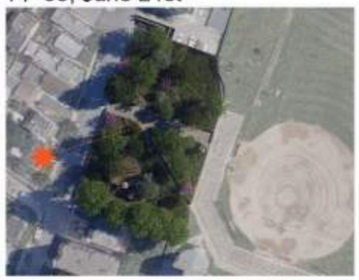
This view highlights the central stepped route through the site. The stair sequence supports daily movement and commuting, while the surrounding terraces create a layered landscape structure. Planting beds, seat walls, and changes in elevation turn the main path into a spatial experience that connects circulation with planted topography.



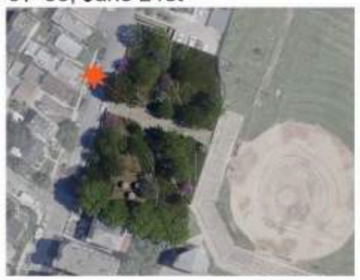
3. Fitness and Social Terrace

This view shows the fitness and gathering area on the terrace. Seating, shade umbrellas, and exercise equipment provide opportunities for rest, social activity, and daily recreation. The surrounding planting and tree canopy help create a shaded and comfortable outdoor room, supporting both active use and casual park life.

7：00, June 21st



9：00, June 21st



11：00, June 21st



13：00, June 21st



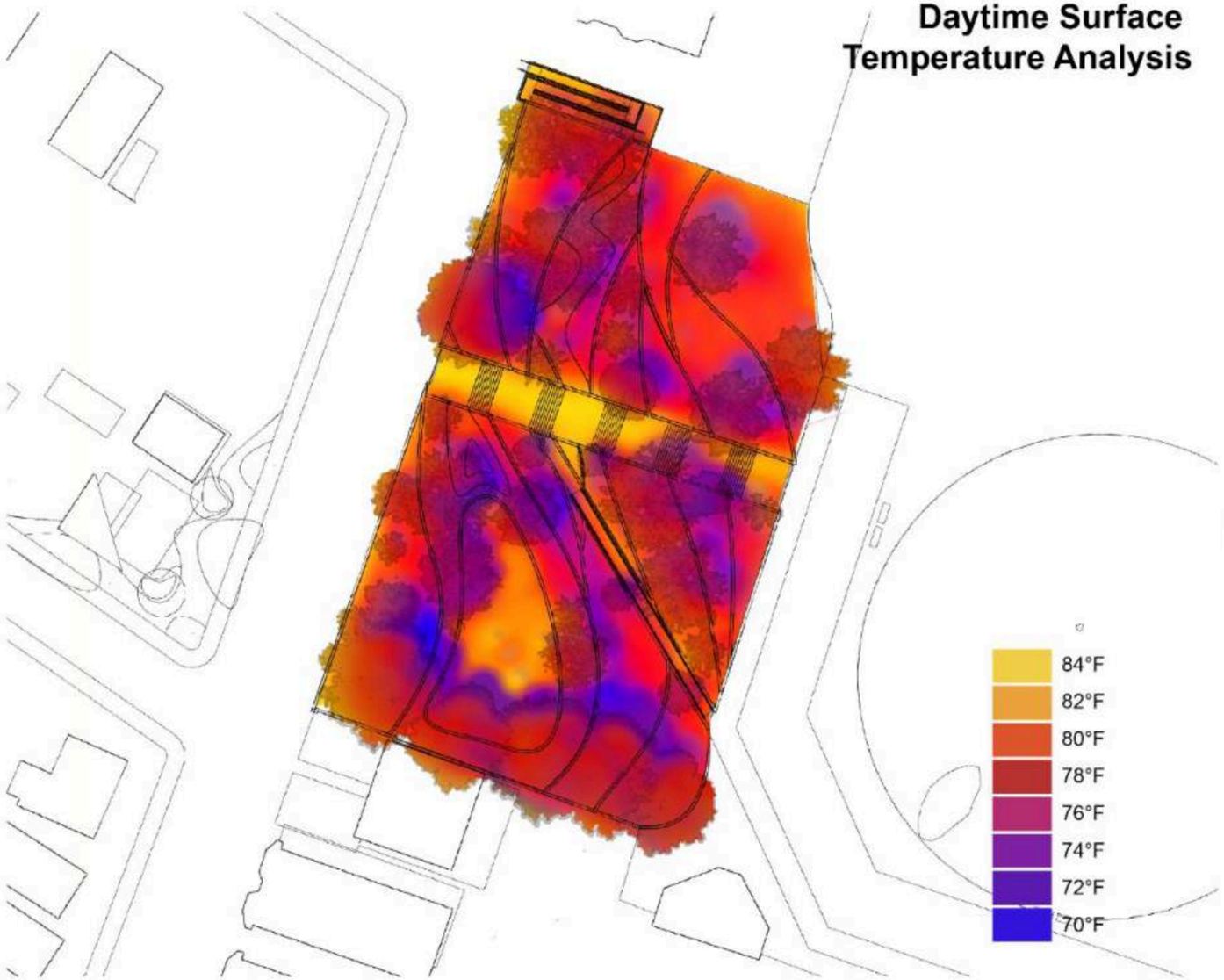
15：00, June 21st



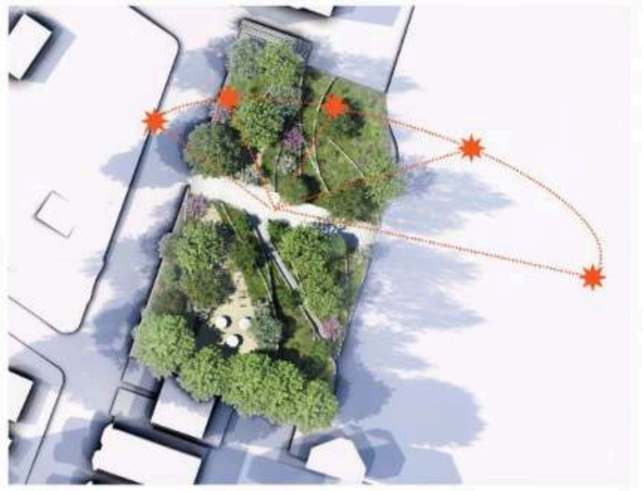
17：00, June 21st



Daytime Surface Temperature Analysis

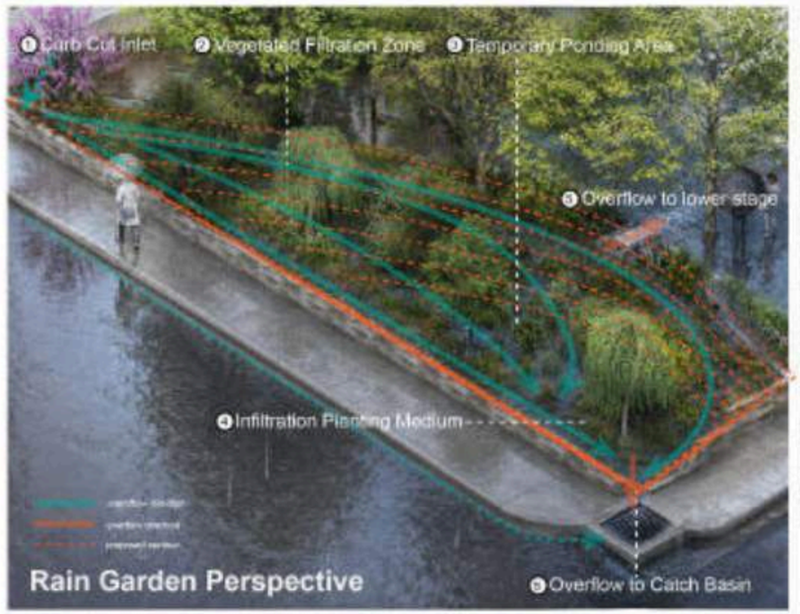


Daytime Shade Analysis



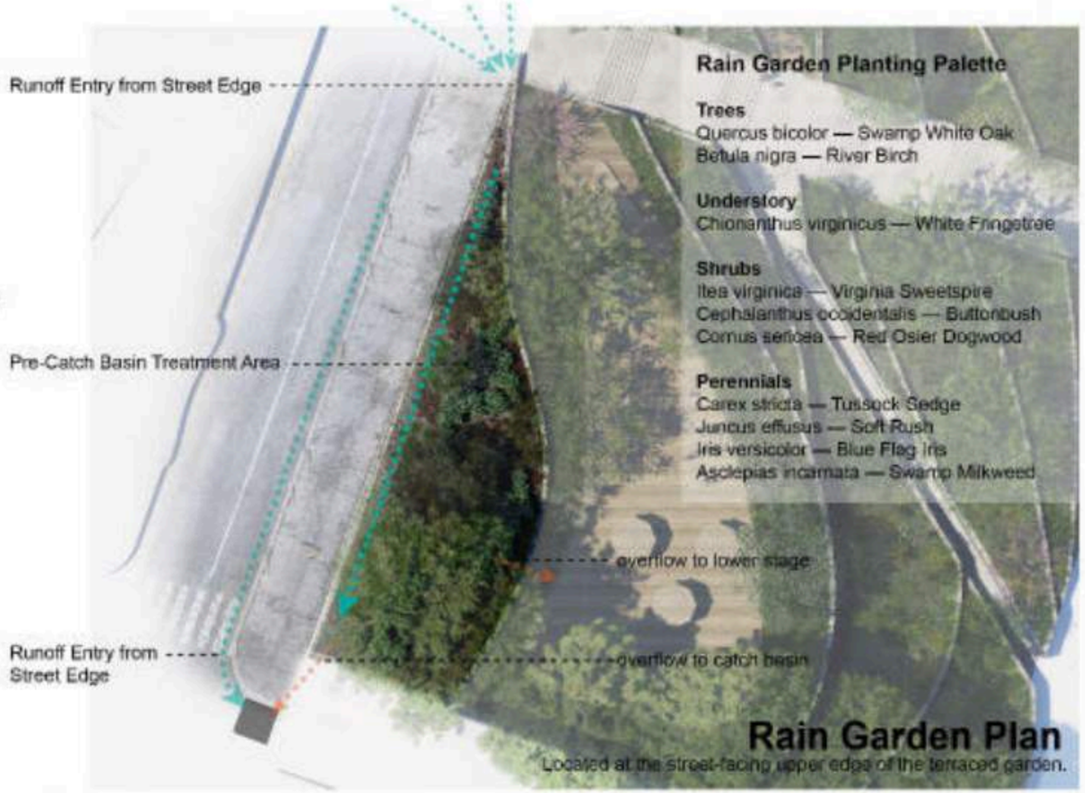
Solar Exposure and Surface Temperature Analysis

This analysis shows how the proposed park node can contribute to urban heat island mitigation. By replacing exposed hard surfaces and underused open ground with layered planting, tree canopy, and shaded pedestrian areas, the design helps lower surface temperature and create cooler microclimates. The June 21st shade and temperature studies show that areas under canopy and dense planting remain noticeably cooler than exposed paved zones.



Terraced Rain Garden as Pre-Catch Basin Storm-water Infrastructure

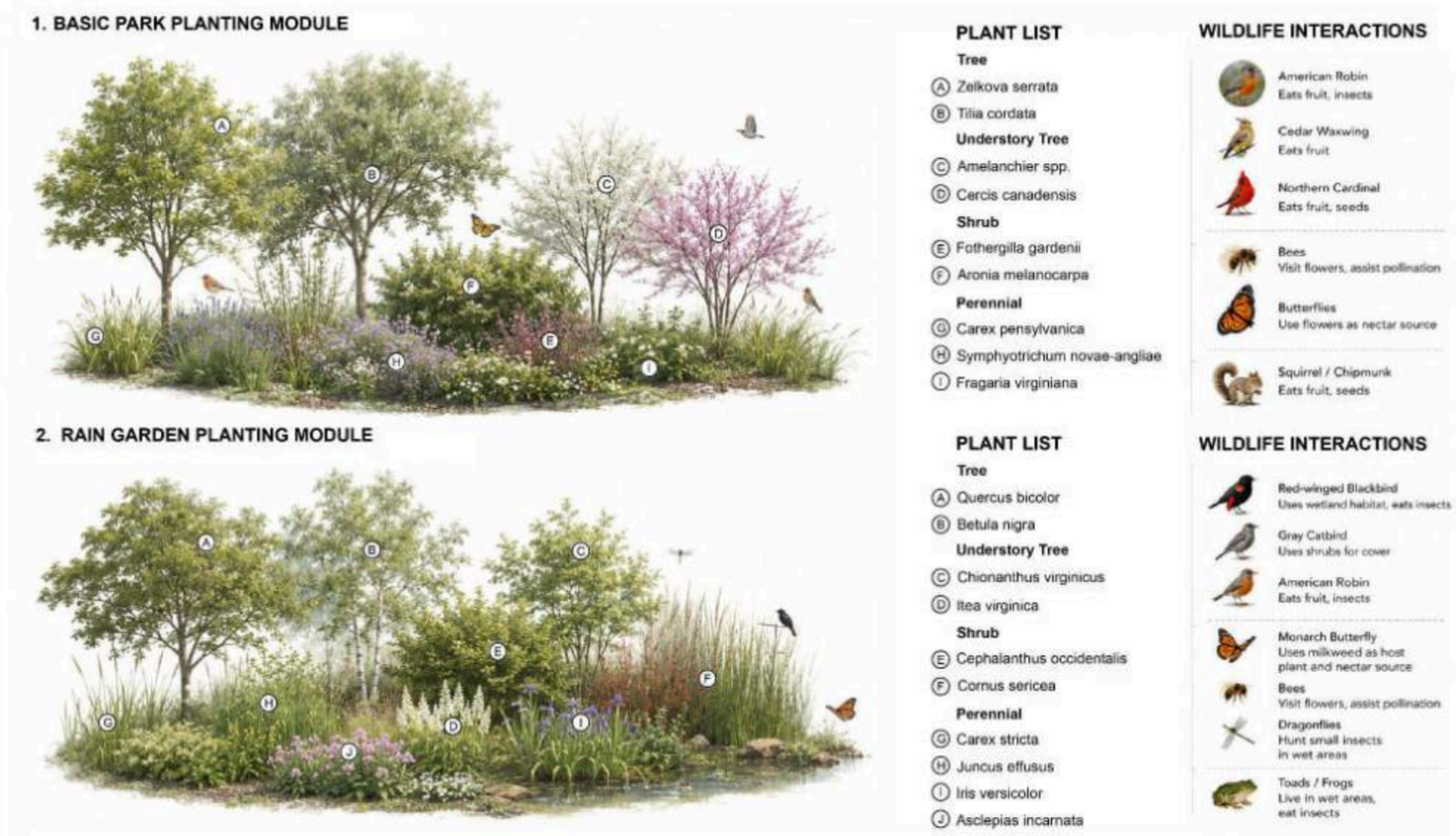
The rain garden planter is located on the upper terrace of the stepped garden, along the street-facing edge, and functions as a pre-catch basin stormwater treatment node. Street runoff first enters through a curb cut, where it is slowed, filtered, and temporarily retained by the planting, soil media, and root zone. Part of the water infiltrates within the upper planter, while excess runoff is gradually distributed downslope through the terraced garden before finally discharging into the catch basin. This system combines stormwater management with layered planting, allowing the rain garden to operate not only as a landscape edge, but also as ecological infrastructure that reduces drainage pressure.







Seasonal sequence of the planting, winter to fall.



Layered planting that shades, holds water, and feeds wildlife

Two modules pair each plant layer with the wildlife it supports. **Canopy trees provide shade**; the **rain-garden module** uses moisture-tolerant species where runoff collects.

Seasonal color and structure of key species: monthly tones, and plant heights from 0 to 2 m.



ReCharge Circuit:

Ecological Remediation & Governance System for Seneca Meadows

2025FA LA6010 Studio_TRANSLOCAL ECOLOGIES
TIANYE WU TW595

2025.12.20

Design Vision

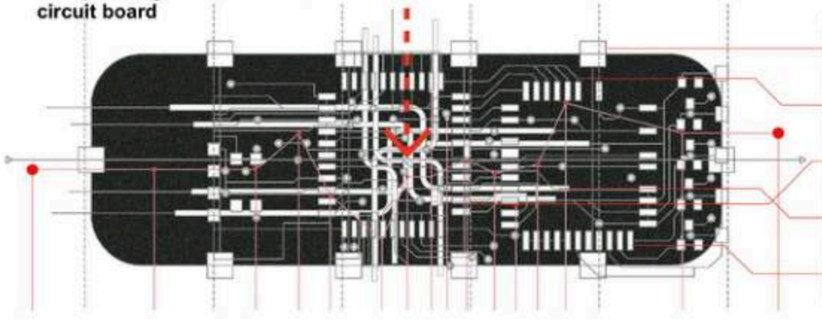
This project conceptualizes the Seneca Meadows Landfill as an "overloaded technological ecosystem." Its environmental challenges—leachate contamination, odor dispersion, slope erosion, and social alienation—are analogized to systemic failures on a circuit board. By introducing the "ReCharge Circuit" landscape framework, the design employs five targeted modular patches as repair components, aiming to transform this degraded industrial site into a monitored, self-healing, and socially equitable public asset.



ISSUE DIAGNOSIS



Morphological reference
circuit board



- ISLAND PATCH
- LINEAR PATCH
- MESH PATCH
- BAND PATCH
- FILTER PATCH

strategy development

CONCEPT GENERATION

Operation & Management

This dimension focuses on the "hardware repair" of the physical site. By analogizing landfill failures to circuit short circuits, modular patches are introduced for intervention. From the initial gravity-driven water filtration (Linear Patch) to slope stabilization (Mesh Patch), the goal is to ultimately achieve an automated, self-healing operational mode for the entire site's functions .

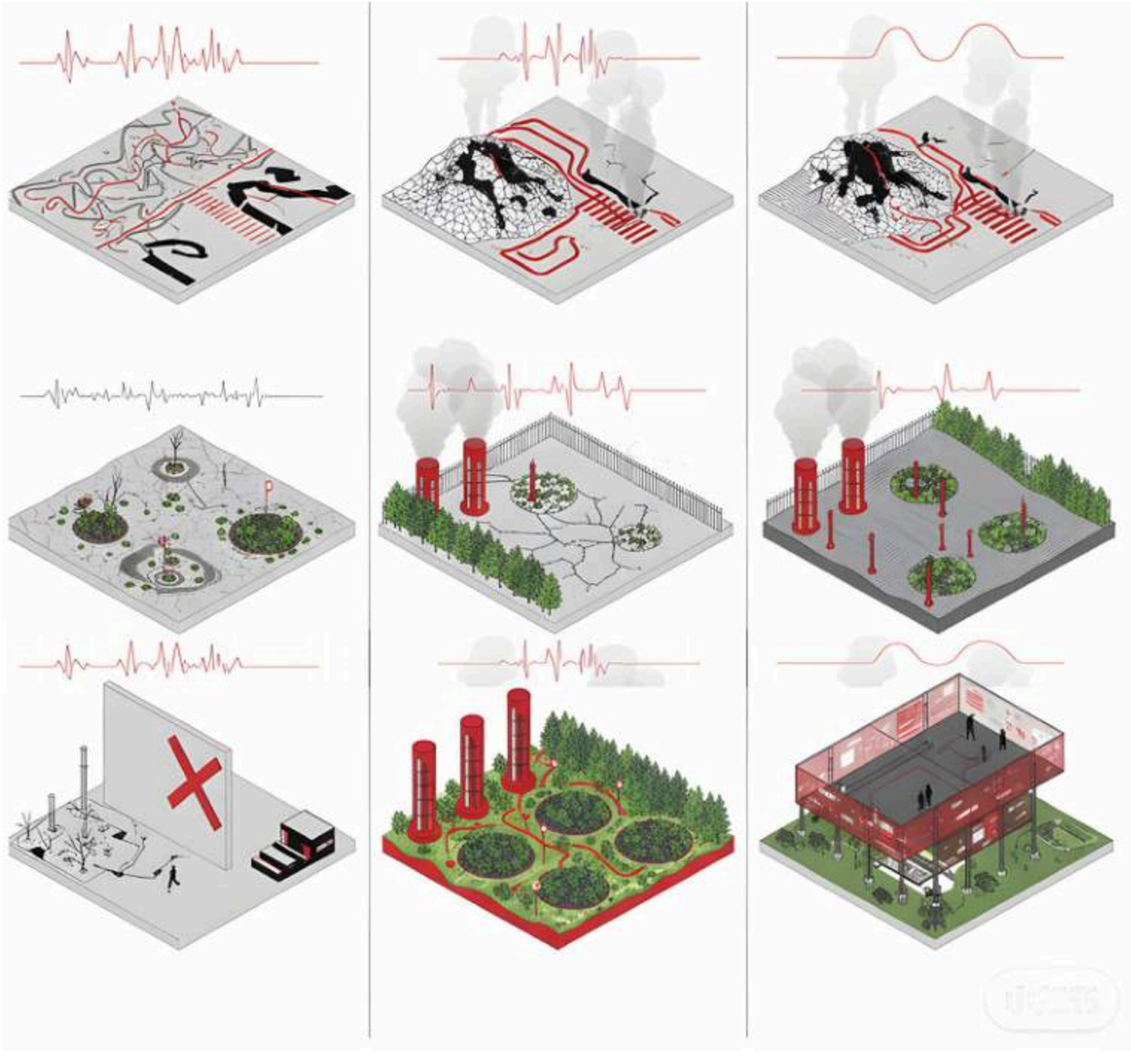
Ecology & Justice

The core lies in the equitable transformation of governance mechanisms. By securing remediation funding through tipping fee surcharges established under the A-6 Rule, closed industrial maintenance is transformed into a visible public good . Restoration outcomes are no longer hidden expenses but aim for social sharing of environmental assets through the gradual transfer of management rights to the community.

Social Cognition

This dimension explores the reconstruction of public perception—specifically, how to break physical and informational walls through "transparent interfaces" . By utilizing real-time data boards and observation platforms, invisible chemical remediation processes are translated into readable scientific narratives, enabling community residents to transition from passive victims to active "citizen scientists" .

Governance Logic



PRESENT

Characterized by "system-wide failures" and information blockages. Uncontrolled leachate infiltration, disordered odor dispersion, and severe slope erosion threaten site safety. At this stage, the landfill is a "NIMBY" black box completely disconnected from the community, with its ecological capacitance in a state of total degradation.

PROCESS

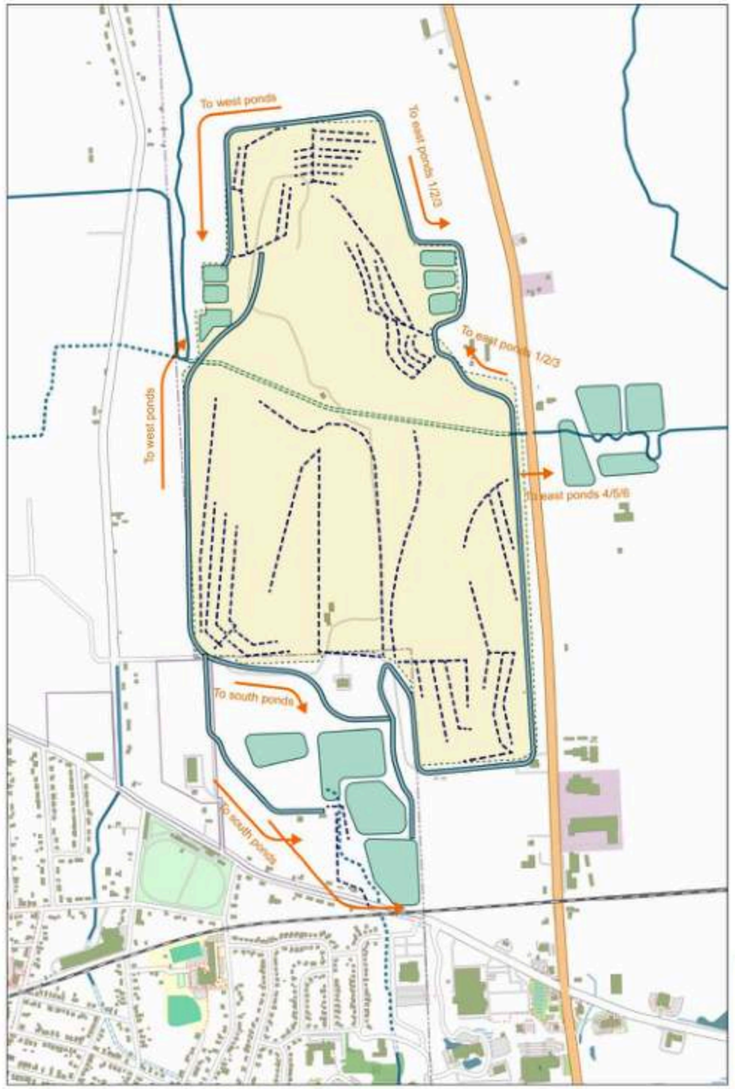
This is the critical stage of patch integration and functional linkage. Various patch modules (Mesh, Linear, Filter) are deployed as needed, with micro-monitoring stations providing real-time feedback on remediation data . It is a process of dynamic balancing, gradually repairing environmental damage and building social trust through technical transparency.

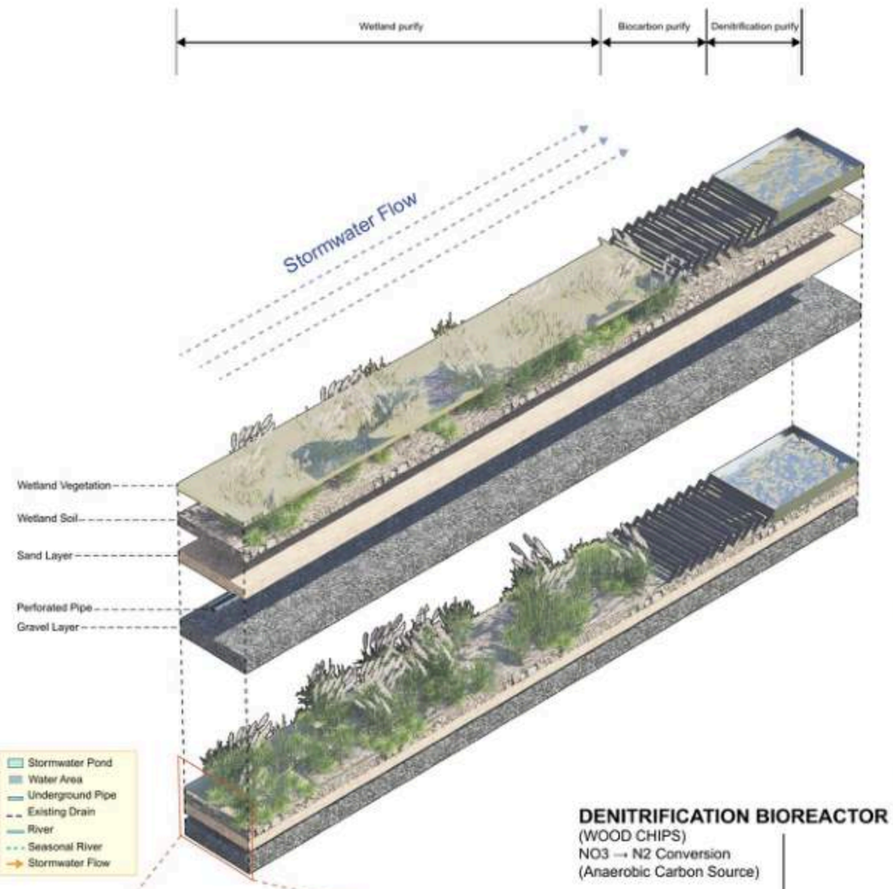
TERMINAL

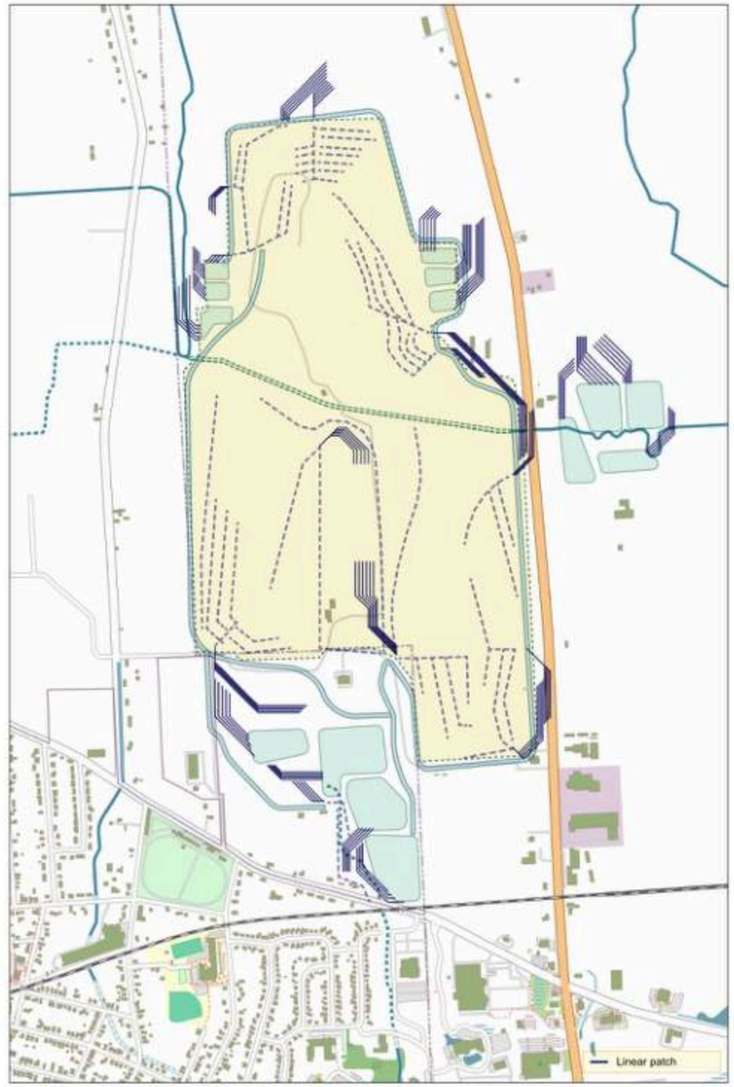
Marks the complete identity shift of the site—evolving from "waste infrastructure" to a "community educational asset." Following the full closure of the landfill, the remediation systems enter a stable self-sustaining period, and management rights are formally transferred to academic and community institutions, serving as a permanent model for ecological education parks.

LINEAR PATCH:
BIOSWALE FILTRATION SYSTEM

The Linear Patch module, the "Circuit Trace" of the "ReCharge Circuit Landscape," addresses leachate contamination and hydrological imbalance. These modular, elongated patches are strategically placed along existing stormwater pond outlets and natural swales. The function is to transform passive discharge points into active biological filtration systems, making the purification process visible and educational.







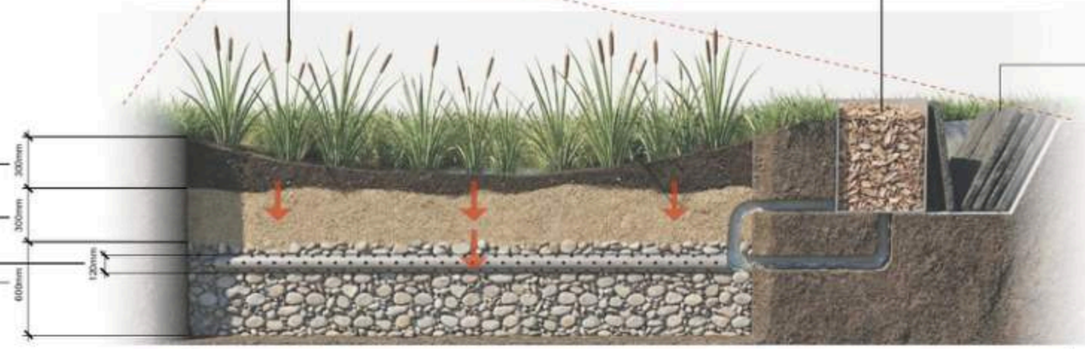
WETLAND VEGETATION
(e.g., Juncus, Carex) – Oxygen Transport & Stabilization
(Nitrification, Bio-active Zone)

WETLAND SOIL
Oxygen Transport & Stabilization
(Nitrification, Bio-active Zone)

SAND FILTER
Microbial Attachment & Fine Filtration
(COD, SS Removal, Nitrification)

PERFORATED PIPE
Flow Distributor
(Pressure Regulation, Flow Buffering)

GRAVEL LAYER
Physical Pre-treatment & Flow Equalization
(Sedimentation, Flow Distribution)



DENITRIFICATION BIOREACTOR
(WOOD CHIPS)
$NO_3 \rightarrow N_2$ Conversion
(Anaerobic Carbon Source)

BIOCARBON CURTAIN
(45° INCLINED PLATES)
Final Adsorption
(Metals, Micro-pollutants, Odor)

Sectional Layer Analysis
: The system features a Top Vegetation Layer for uptake and stabilization, a Filter Media Layer (gravel/woodchip/biochar), an Impermeable Liner at the base for controlled flow, and an Underdrain System for treated effluent collection.

Feasibility & Site Application

Strategic Placement: Placement at Stormwater Pond Outlets and Existing Swales utilizes existing drainage infrastructure, minimizing new earthwork and leveraging the low-lying basin topography for energy-efficient, gravity-driven treatment.

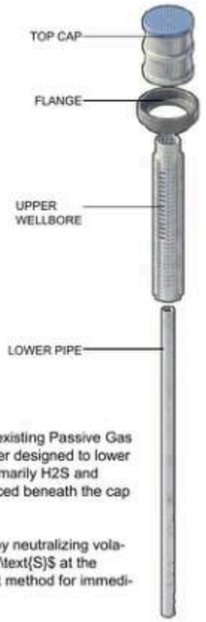
Sustainability & Cost: The modular design (20-50m segments) allows for phased, low-cost construction and easy replacement of media. Use of local materials (woodchips, biochar) enhances sustainability and community collaboration.

Risk Mitigation: Bypass pipes are incorporated to maintain function during winter/low-efficiency periods. Liner systems prevent re-infiltration into groundwater.



FILTER PATCH:
GAS MANAGEMENT & ATMOSPHERIC CONTROL

The Filter Patch module is designed to address the primary community conflict at Seneca Meadows Landfill: odor dispersion and atmospheric pressure issues caused by methane (CH_4) and hydrogen sulfide (H_2S). Conceptually, it functions as the "Leaking Socket" repair module in the "ReCharge Circuit Landscape," aiming to make the invisible subsurface gas processes manageable and readable by the public. These patches are a strategic combination of low-tech filtration devices and biological windbreaks, transforming emission points into controlled, visually integrated elements that actively buffer the community from nuisance.



ACTIVATED CARBON CAP
(Ø1.2m, H=0.4m)
Perforated Top, Replaceable Filter Unit



Carbon Cap (Point Filtration):

Structure: Activated Carbon Filter installed on existing Passive Gas Wells. The cap is a modular, replaceable canister designed to lower the out-venting of odor-causing compounds, primarily H2S and thiols. Condensate collection sumps are reinforced beneath the cap to manage internal well pressure.

Function: Directly addresses odor complaints by neutralizing volatile organic compounds (VOCs) and H_2S at the emission source. This is a low-cost, high-impact method for immediate relief.

ZONE 1: LOW SHRUBS

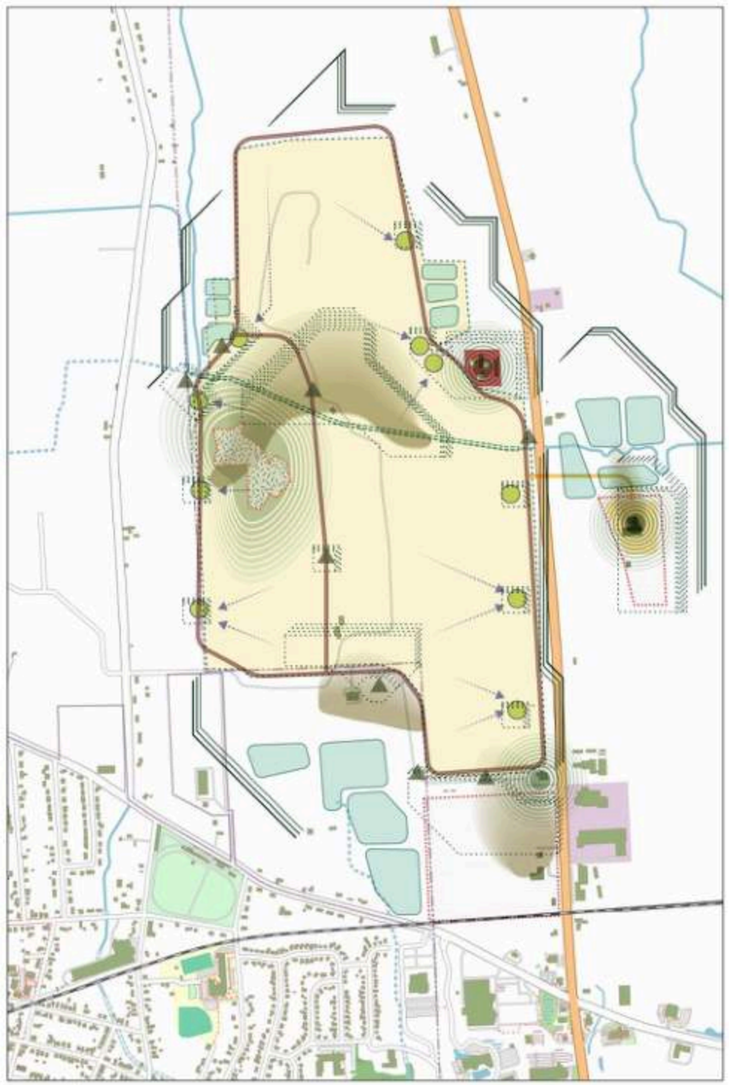
(e.g., Redosier Dogwood, Ninebark)
Function: Trap low-altitude dust & particulates;
reduce near-ground wind speed.

ZONE 2: FAST-GROWING DECIDUOUS TREES

(e.g., Eastern Cottonwood, Hybrid Willow)
Function: Create turbulence to dilute odors; rapidly

ZONE 3: TALL EVERGREENS

(e.g., E. Red Cedar, White Spruce)
Function: Year-round wind blocking & visual screen; permanent structure.
establish barrier height.



Stormwater Pond
Water Area
Underground Pipe
Existing Drain
River
Seasonal River
Stormwater Flow

Windbreak Hedges (Boundary Buffering):

Structure: Dense, multi-row plantings of non-invasive native coniferous/broadleaf mixed trees and shrubs along the site perimeter, particularly on the downwind sides and wind corridors.

Function: Physically reduces near-ground wind speed, causing odor plumes to mix and lift, aiding dispersion before reaching the community. The foliage also acts as a biological filter for dust and residual aerosols.



Populus deltoides

Salix x matsudana x alba



Cornus sericea



Physocarpus opulifolius

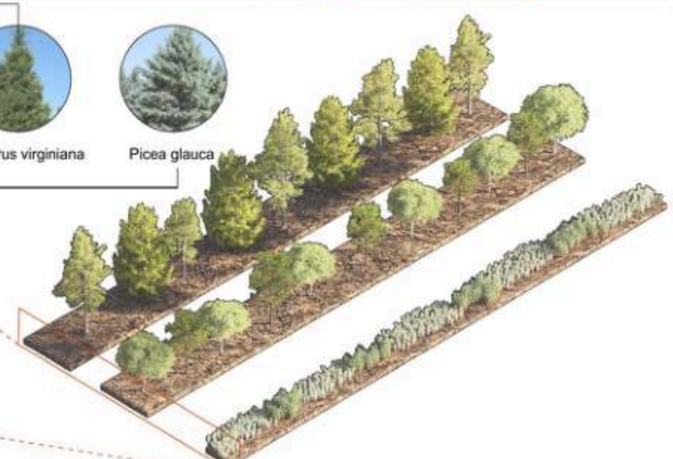
Wind & Odors Direction





Juniperus virginiana

Picea glauca



BAND PATCH
Social Interface & Cognitive Reconstruction

Within the "ReCharge Circuit Landscape," the Educational Park Nodes function as the "Port" for repairing "Disconnected Ports," serving as the only safe, continuous, and interactive interface for the community. They transform passive site visits into active learning and civic science participation.



Educational Park Node

Social & Ecological Interface, Knowledge Transfer
Access & Boundary & Experimentation





Real-time Data Boards

Digital or updated physical display boards located at Observation Points.

Transparency: Visually tracks Odor Mitigation (H2S, wind direction vs. complaints) and Water Remediation Progress (Linear Patch KPI tracking).



Observation Points

Structure : Elevated wooden observation platforms installed outside the main fence line for safety compliance, yet providing an unimpeded view of the Landfill's engineered surface.

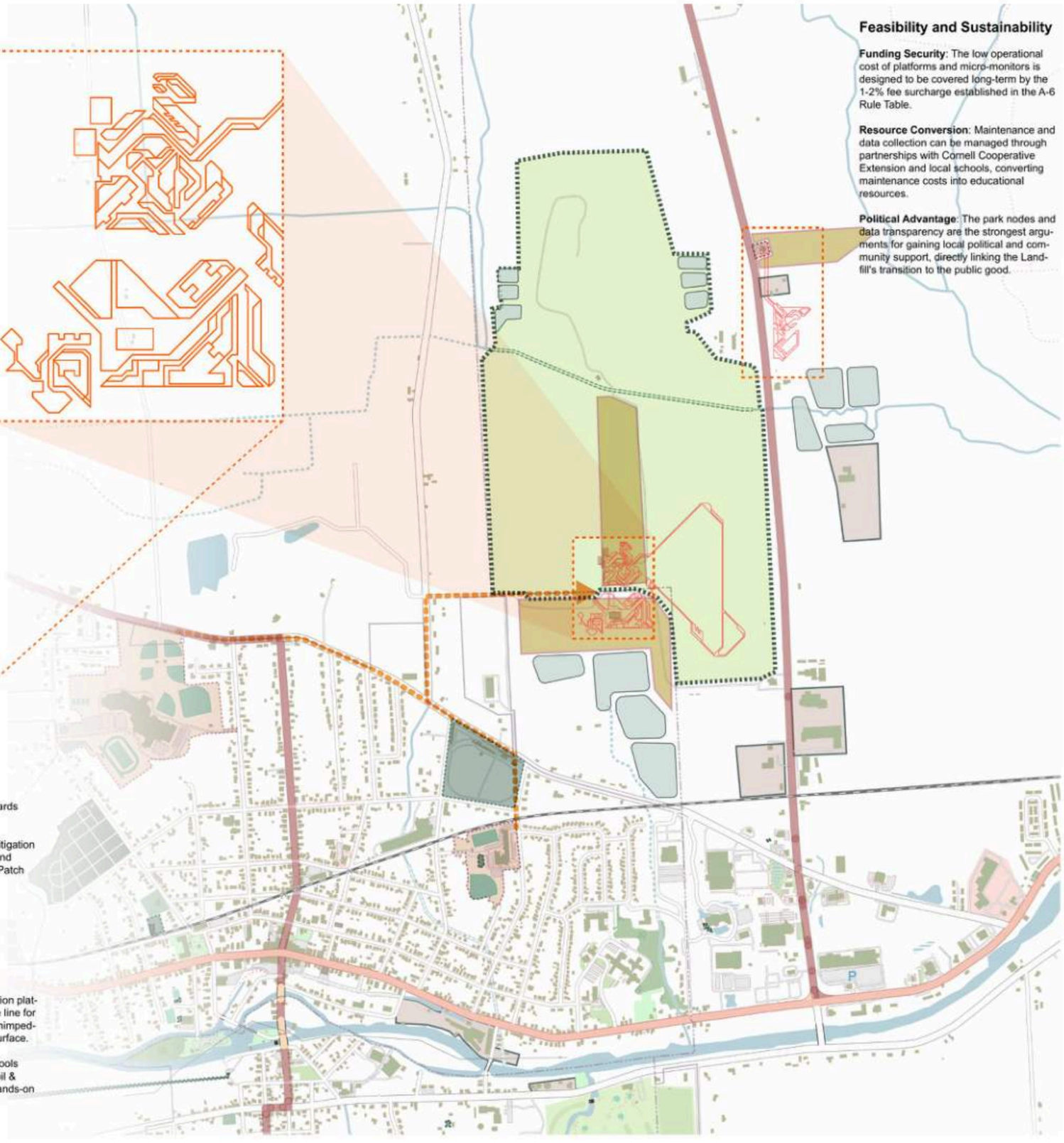
Targeted Application: Nodes near schools should emphasize Island Patch for Soil & Microbial Restoration to encourage hands-on student involvement.



Micro-monitoring Stations

Low-cost micro-sensors for air quality (odor/wind direction), temperature, and water sampling points for Linear Patch effluent.

Citizen Science: Provides raw, accessible data for community/school monitoring projects.



Feasibility and Sustainability

Funding Security: The low operational cost of platforms and micro-monitors is designed to be covered long-term by the 1-2% fee surcharge established in the A-6 Rule Table.

Resource Conversion: Maintenance and data collection can be managed through partnerships with Cornell Cooperative Extension and local schools, converting maintenance costs into educational resources.

Political Advantage: The park nodes and data transparency are the strongest arguments for gaining local political and community support, directly linking the Landfill's transition to the public good.

MESH PATCH:
SLOPE STABILITY & EROSION CONTROL

The Mesh Patch is the "Engineering Skin" of the "ReCharge Circuit Landscape," addressing the critical risk of slope erosion and instability on the 400-foot landfill mound. Conceptually, it acts as the "broken trace repair" module, stabilizing the physical geometry. Placed on high-stress slopes, the patch utilizes geotechnical reinforcement to create a stable, maintainable foundation for long-term ecological growth.

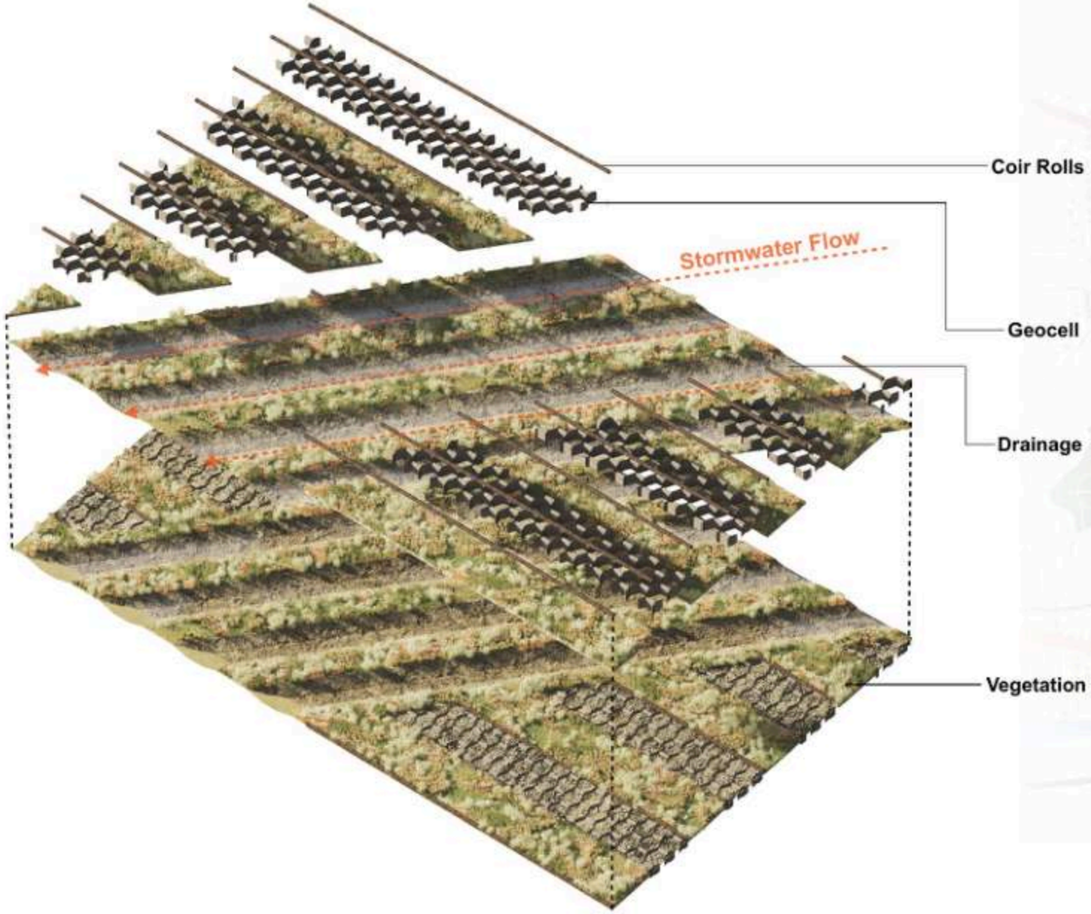
Structural Mechanism

The patch is a geotechnical-ecological hybrid system designed for managing shear stress and runoff.

Lateral Confinement (Geocell): Hexagonal Geocells (made of dark aggregate material) are laid onto the cap to provide lateral confinement to the cover soil, preventing sliding and distributing the load of differential settlement across a wider area.

Surface Control (Coir Rolls & Drainage): Coir Rolls are integrated along contours to act as check dams, slowing surface runoff and trapping sediment. Perforated pipes and trench drains at the base rapidly channel water away, preventing saturation and slope failure.

Dynamic Anchoring (Vegetation): Erosion-resistant native grasses are established within the Geocell cells, ensuring that as the Coir Rolls biodegrade, the natural root system takes over the binding function, making the stability self-sustaining in the long term.



Active Site

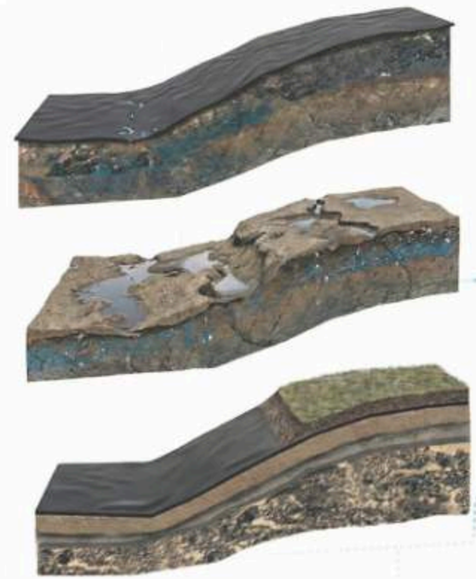
Daily cover is thin and loose. Rainwater infiltrates vertically directly into the waste mass, generating significant leachate.

Final Covered

Thicker soil cover promotes some lateral flow. However, without an impermeable liner, rainwater still penetrates through to the subsurface.

Capped Area

An engineered system with an impermeable liner (black line). It completely blocks infiltration, forcing all water to drain laterally, isolating the waste mass entirely.





Feasibility and Application

The Mesh Patch is an essential and cost-effective long-term solution for SML's unique challenges.

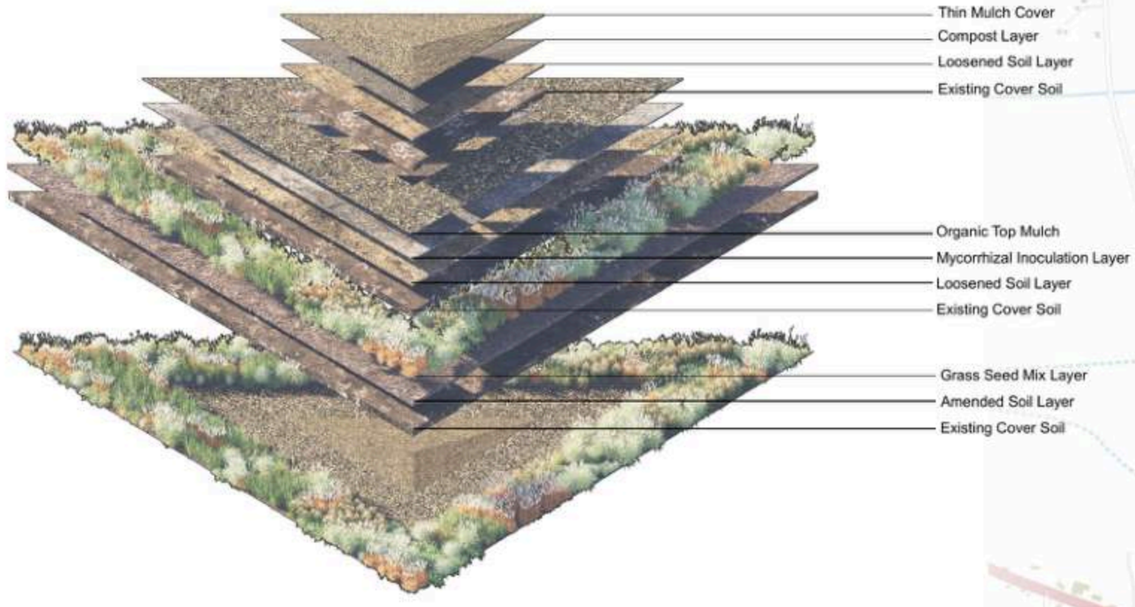
Strategic Focus: Patches are only placed on high-stress slopes prone to erosion and shear stress (as shown in your plan), maximizing the impact of investment.

Sustainability: The method significantly reduces future remedial erosion repair costs. The geometric Mesh Texture serves as the engineered foundation layer, aesthetically signaling the site's transition from waste dump to a structured ecological base.

Aesthetic Integration: The visual texture of the mesh symbolizes the engineered foundational layer beneath the recovering ecology, embodying the principle: "Engineering + Ecology Skin."

ISLAND PATCH:
Soil & Microbial Regeneration

The Island Patch module is conceptualized as the "Degraded Capacitor" repair unit within the "ReCharge Circuit Landscape". Its primary purpose is to address the core problem of soil fatigue and microbial loss on the landfill cap. Strategically implemented as small, modular plots, the patch embodies the philosophy: "Regeneration starts locally," transforming the biologically sterile cover into a vibrant, self-healing ecological substrate.



Action: The existing cover soil is lightly tilled and a layer of Native Compost is applied. The plot boundary is clearly defined.

Function : This phase initiates organic matter renewal and immediately improves water retention and infiltration.

Phase 1: Compost Cap Zone

Action: An Organic Top Mulch layer is applied, and the soil is treated with a Mycorrhizal Inoculation Layer. This is often done by spot-application or using local soil washing to introduce beneficial fungi.

Function: This is the core "Regeneration" step, boosting microbial activity and enabling plants to access nutrients and water more efficiently, laying the groundwork for plant growth.

Phase 2: Mycorrhizal Inoculation Zone

Action: The final layer of Amended Soil (with compost) is applied, followed by seeding of Native Grass and Non-Invasive Forbs. This layer is actively monitored by High School/College Volunteers for Vegetation Cover and Soil Respiration.

Function: Establishes permanent surface stabilization and the root structure necessary for long-term ecological self-sufficiency, embodying the philosophy: "Regeneration starts locally".

Phase 3: Grass Mix Zone





SOIL
TREATMENT OVER TIME



GRASSLAND

Site Role: Acts primarily as the "Engineering Skin" (Mesh Patch) for the 400-foot mound slopes and the cover layer for initial soil improvement units (Island Patch).

Successional Timeline: During Years 1–5, 15cm of native compost and mycorrhizal inoculation are applied to rapidly awaken the activity of sterile cover soil . Native herbaceous plants utilize dense root systems to establish initial nutrient cycling.

Ecological Benefits: Provides immediate erosion control and prevents slope slippage through the "dynamic anchoring" of vegetation roots, creating a water-retaining and breathable substrate for subsequent woody plant colonization.

WOODLAND

Site Role: Strategically placed along the landfill perimeter (Filter Patch) and on both sides of hydrological paths (Linear Patch), serving as an ecological interface between the site and the community.

Successional Timeline: Between Years 5–15, a multi-layered structure is formed by fast-growing deciduous broadleaf forests (e.g., Hybrid Willow, Eastern Cottonwood) and tall evergreens (e.g., Red Cedar, White Spruce) . The understory begins to develop, increasing habitat complexity.

Ecological Benefits: Acts as a "controlled breathing" biological barrier, physically intercepting dust and utilizing turbulence to dilute and neutralize odorous gases (e.g., H_2S), significantly reducing negative environmental perceptions of the site121212121212121212.

LOWLAND FOREST

Site Role: Forms in low-lying basins (Lowland Forest) and at the outlets of linear filtration systems, serving as the ultimate ecological receptor for the site.

Successional Timeline: Represents the 15–40 year long-term successional climax. As soil capacitance is fully restored and water cycles deeply, the habitat evolves into a mature biomatrix with the most stable structure and highest species richness.

Ecological Benefits: Achieves final purification of surface runoff and long-term biodiversity support, marking the formal handover and return of site management from industrial operators to community assets.

Liquid Legacy

The Revival of Zhonghua Shipyard

Landscape Architecture Design
Location : Shanghai , China
Time : 2024.02
Type : Individual Work
Instructor : Kayden Xu

Liquid Legacy transforms the historic Zhonghua Shipyard into a wastewater remediation park. Combining heritage preservation with ecological restoration, the site narrates water's journey from pollution to renewal through interactive AR experiences, revitalized industrial structures, and lush wetlands, fostering public awareness of sustainable water management and environmental recovery.



Location Analysis



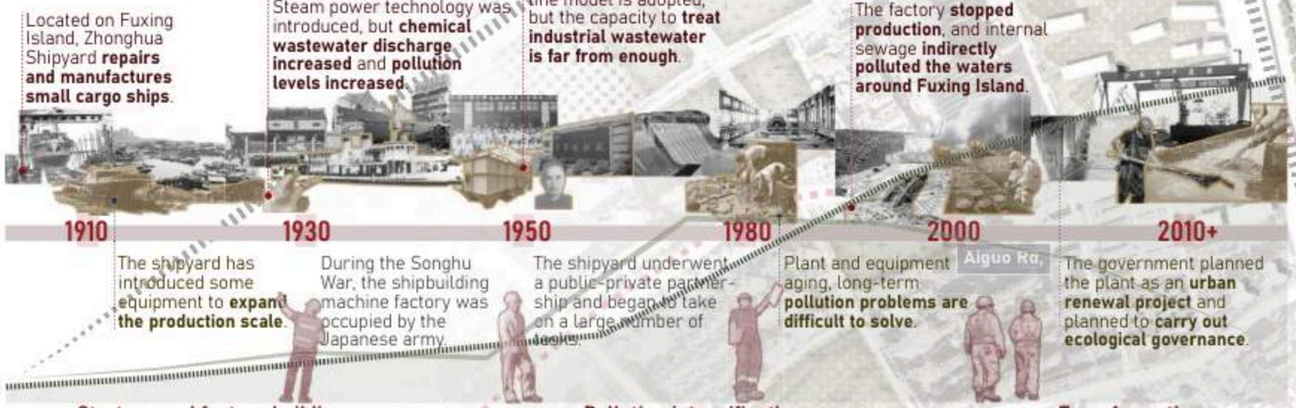
“ Shanghai industrial wastewater discharge **312.17 million tons**

It accounts for **15.8%** of the total wastewater ”

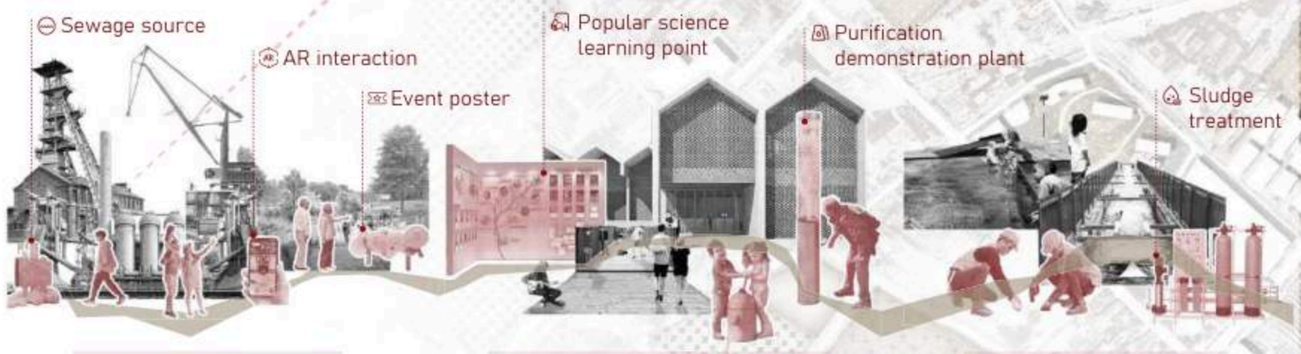
The content of petroleum pollutants **226 Tons**

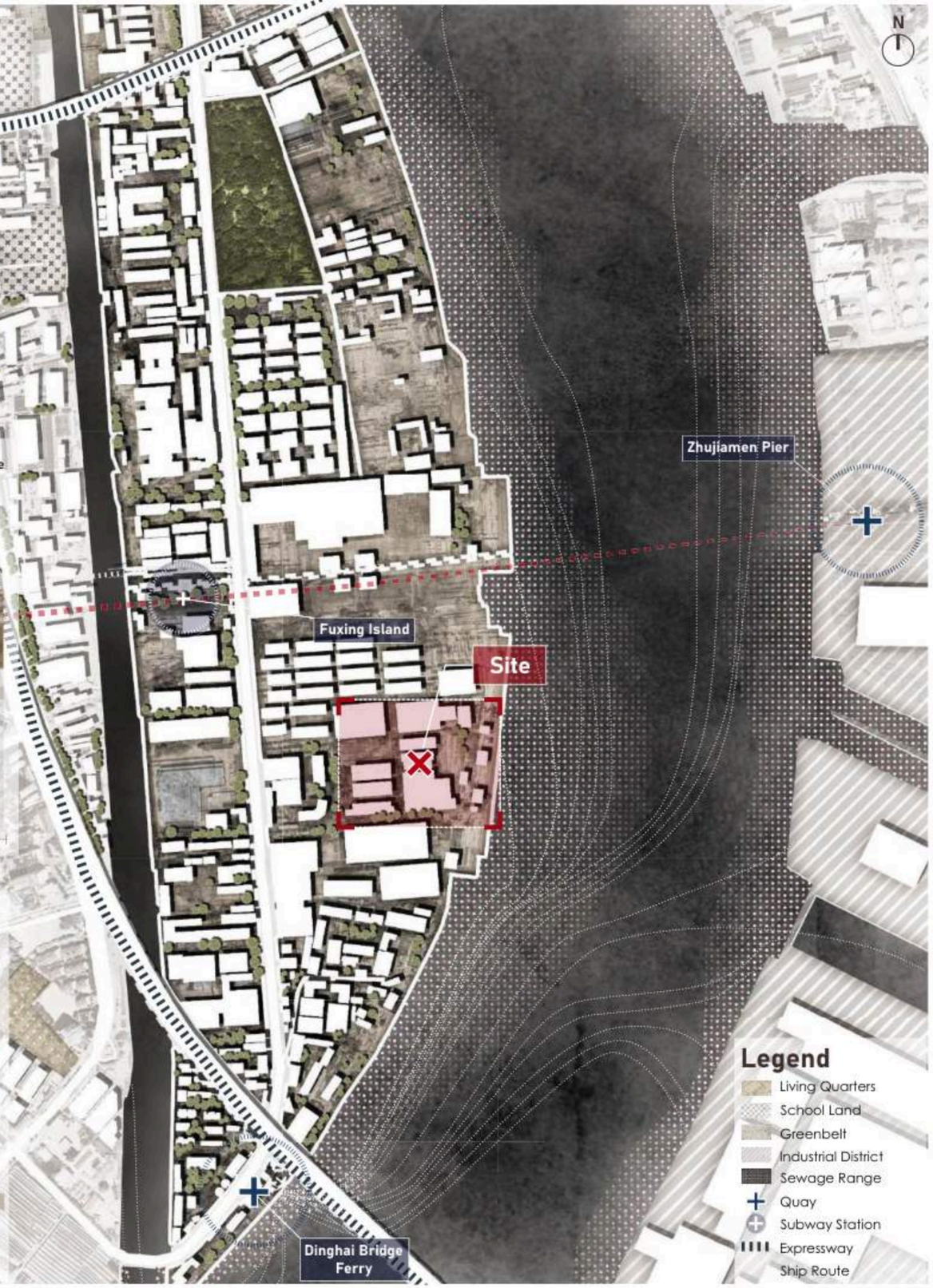
The content of petroleum pollutants **239 KG**

History

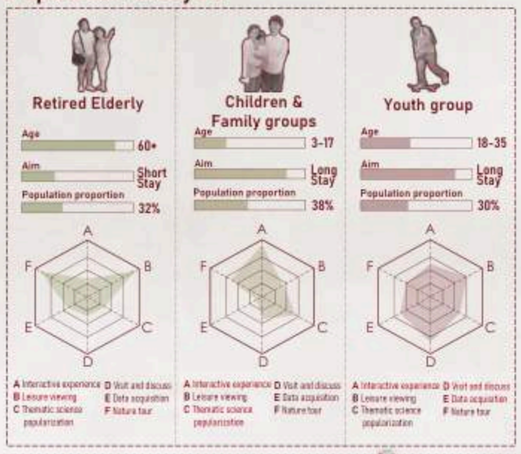


Overall design strategy

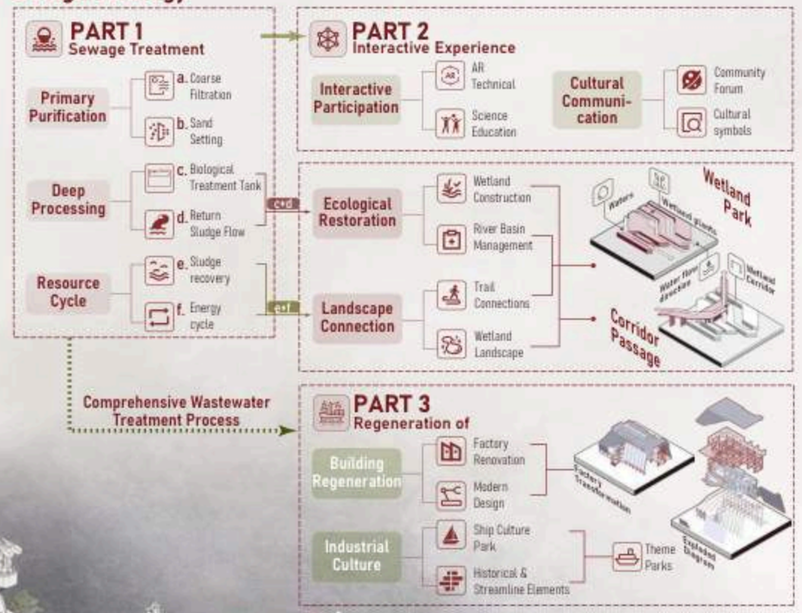




Population Analysis



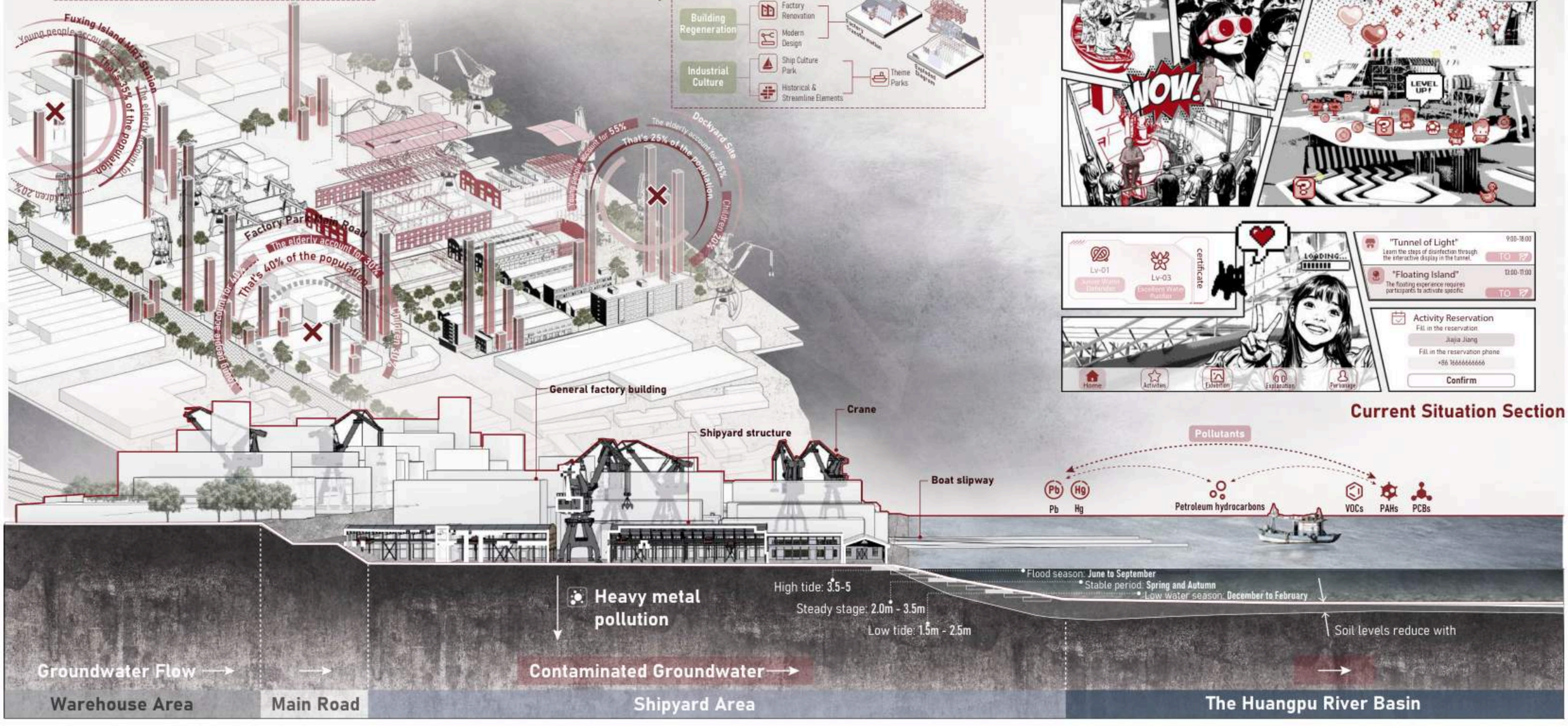
Design Strategy



Event Story Line

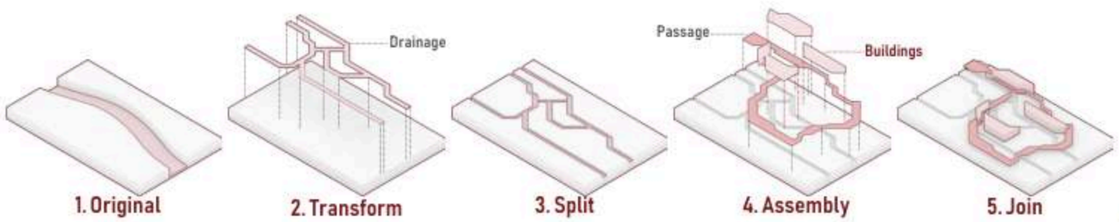


Current Situation Section



Shape Generation

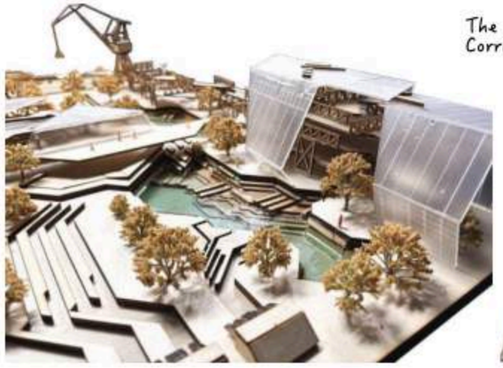


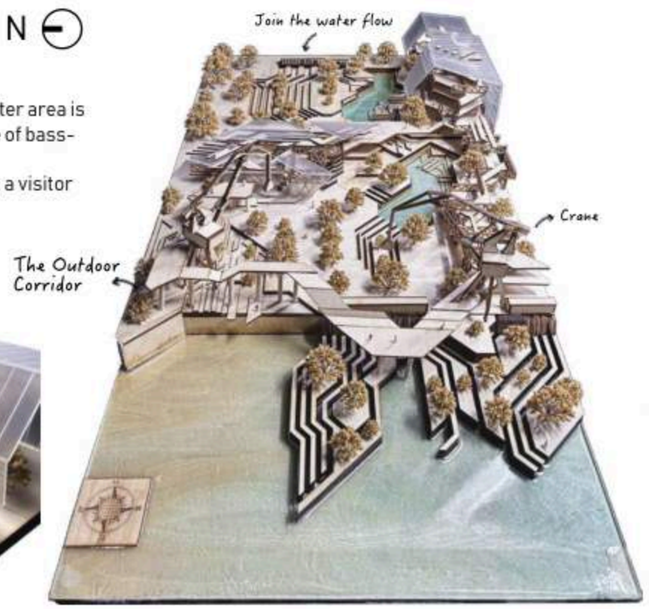


Physical Model

Size: 900mm*600mm
Material: Basswood board, Acrylic sheet, Resin
Scale: 1:300

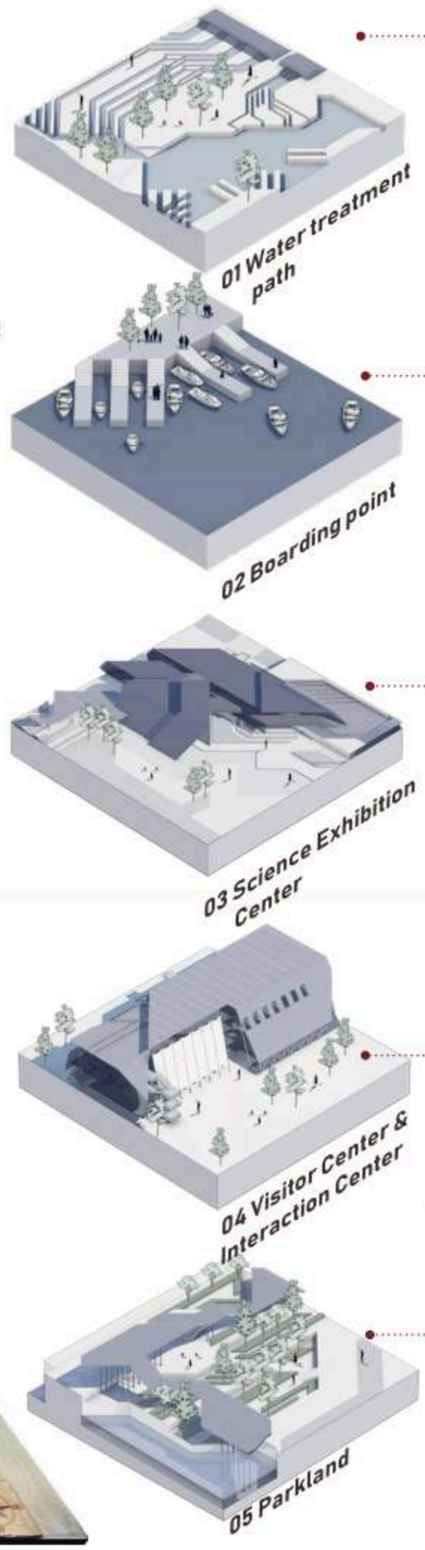
The factory building is made of acrylic panels, the water area is filled with resin, and the rest of the structure is made of bass-wood.
This hand-made model includes an outdoor corridor, a visitor center and a science center.







Node Axonometric



Master Plan

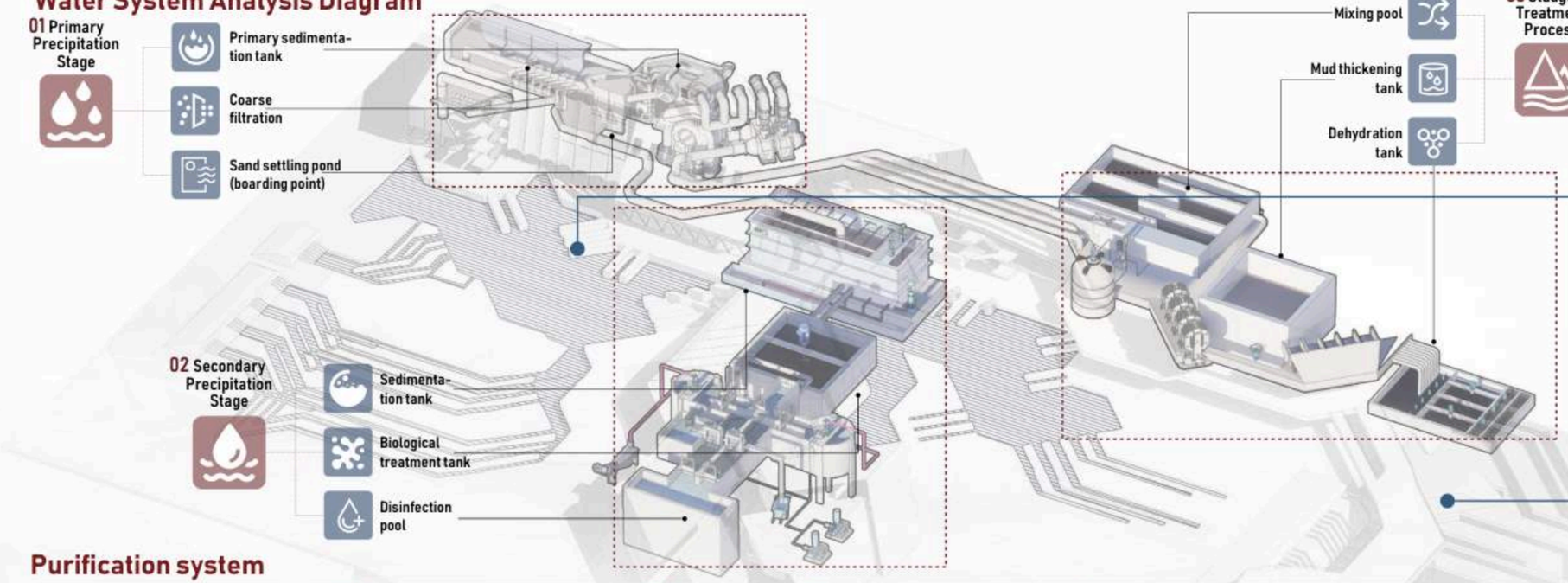


Water System Analysis Diagram

01 Primary Precipitation Stage
- Primary sedimentation tank
- Coarse filtration
- Sand settling pond (boarding point)

02 Secondary Precipitation Stage
- Sedimentation tank
- Biological treatment tank
- Disinfection pool

03 Sludge Treatment Process
- Mixing pool
- Mud thickening tank
- Dehydration tank





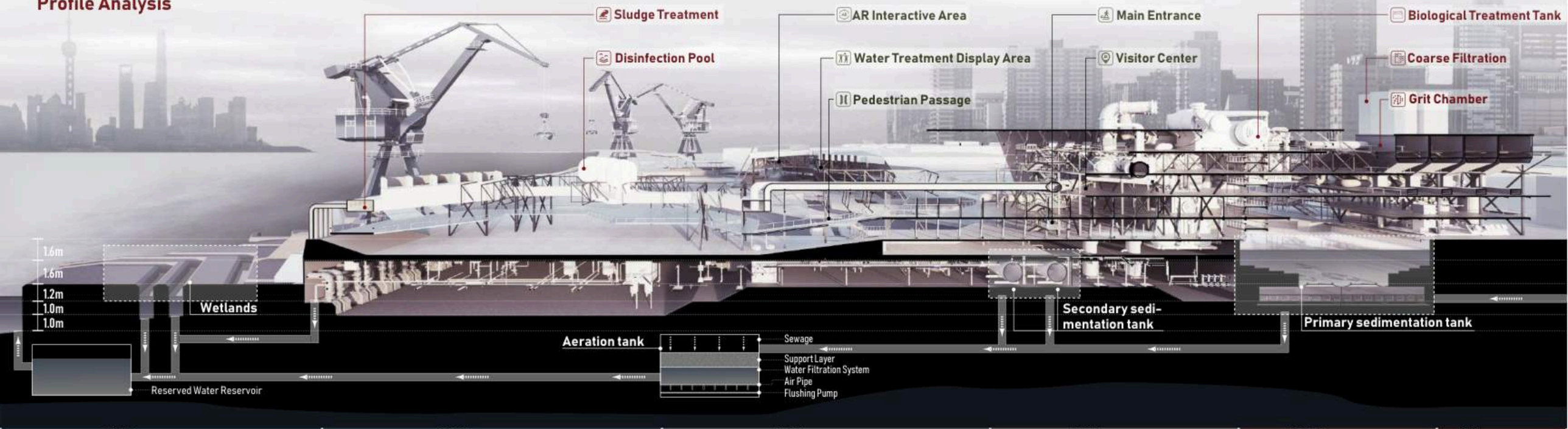


Purification system

Municipal sewage → Coarse filtration → Sand settling pond → Primary sedimentation tank (01 Primary Precipitation Stage) → Biological treatment tank → Sedimentation tank (02 Secondary Precipitation Stage)

- Filter out the sludge → Mixing pool → Mud thickening tank → Dehydration tank (03 Sludge Treatment Process) → made into fertilizer
- Dosing → Mixing pool
- filter → Disinfection pool → disinfect → produce clean water → For wetlands

Profile Analysis



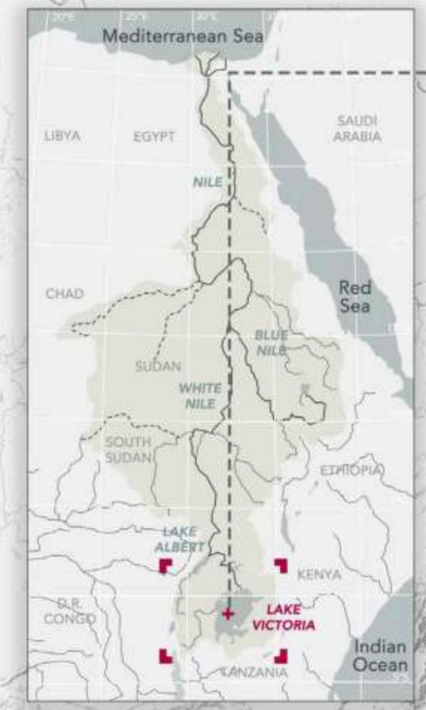
95% purification	80% purification	60% purification	40% purification	20% purification	0% purification

Where Natural Sustainability Is Visualized

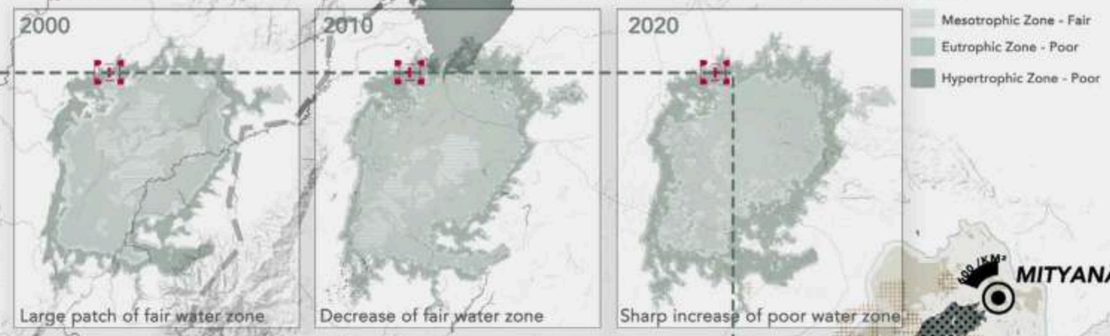
Location: Lake Victoria, Africa
Duration: 2024.6-2024.8
Software: GIS, Photoshop, Illustrator, Rhino
Individual Work
Climate Resiliency

Lake Victoria is one of the African Great Lakes.
The long-term human intervention has continuously disrupted the ecological security pattern of Lake Victoria, leading to floods, water source pollution, damage to the fishery economy, wetland degradation, and so on.
Thus new interventions are introduced to establish an interconnecting and stable relationship between the human and the Lake Victoria hydrological system.

THE NILE RIVER BASIN



EUTROPHICATION TREND IN LAKE VICTORIA *(2000-2020)*

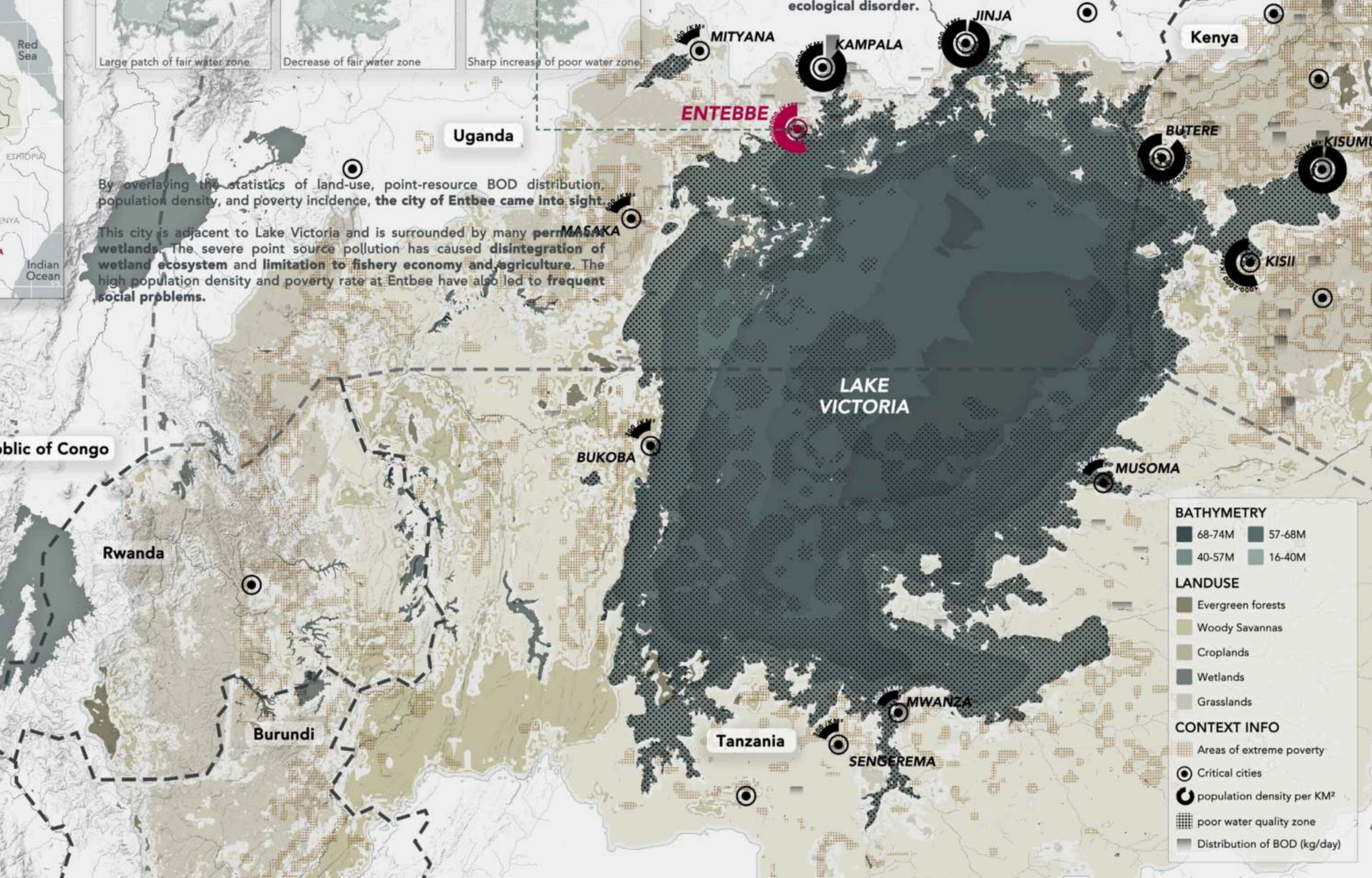


Lake Victoria is one of the **African Great Lakes**, with a surface area of 59,947km². It is home to many species of fish which live nowhere else. The lake's area is divided among three countries: **Tanzania, Uganda, and Kenya.**

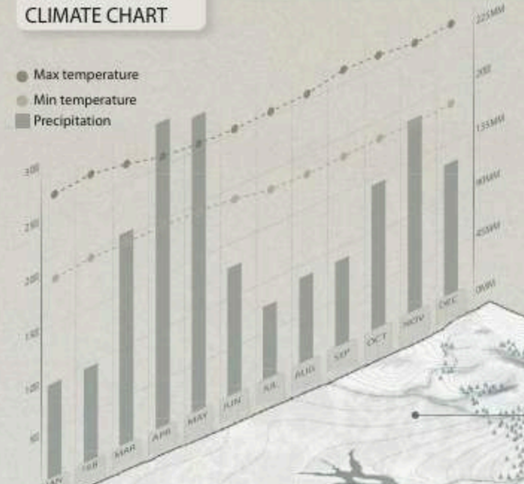
With a solid research in the context of Lake Victoria Watershed, it shows that there's a **severe phenomenon of eutrophication**, the water quality is getting worse sharply which leads to **aquatic ecological disorder.**

By overlaying the statistics of land-use, point-resource BOD distribution, population density, and poverty incidence, **the city of Entbee came into sight.**

This city is adjacent to Lake Victoria and is surrounded by many **permanent wetlands**. The severe point source pollution has caused **disintegration of wetland ecosystem** and **limitation to fishery economy and agriculture**. The high population density and poverty rate at Entbee have also led to **frequent social problems.**



ENTEBEE CLIMATE CHART



HYDROLOGICAL ISSUES ON SITE

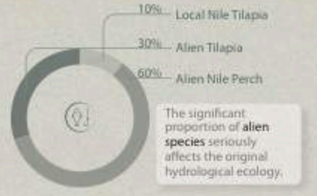
- INUNDATION
- WETLAND DEGREDATION
- BIOLOGICAL INVASION
- EUTROPHICATION

WETLAND ECOLOGICAL ISSUE

ECOLOGICAL SECURITY PATTERN + SOCIAL SECURITY STRUCTURE

FISHING ECONOMIC ISSUE

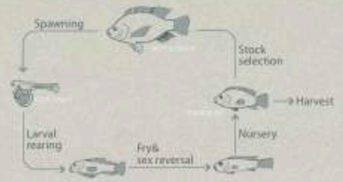
PROPORTION OF FISH CATEGORIES IN LAKE VICTORIA



The significant proportion of **alien species** seriously affects the original hydrological ecology.

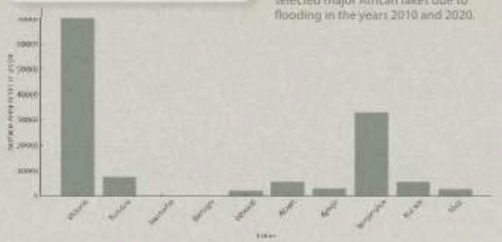
LIFE CYCLE OF TILAPIA

Nursery/Hatchery: 2-3 months
Grow-out Fishing: 4-8months

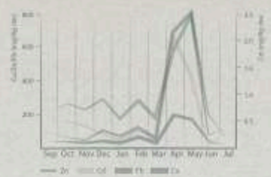


FLOODING AFFECTED RANKING OF LAKE IN THE AFRICA

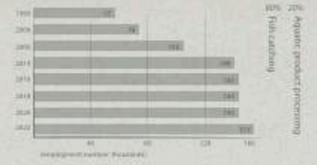
Calculated areal extent (km2) and percentage change in surface area of the selected major African lakes due to flooding in the years 2010 and 2020.



DISTRIBUTION OF HEAVY METAL IN LAKE VICTORIA



EMPLOYMENT OF FISHING INDUSTRY IN UGANDA



FISH DISTRIBUTION IN LAKE VICTORIA



MAJOR FISH TYPE IN LAKE VICTORIA

Lake Victoria in Africa hosts diverse fish species, including Nile perch and tilapia. Distribution varies by depth and region, influenced by water temperature, oxygen levels, and ecological conditions.



NATURAL CONDITION

HUMAN INTERVENTION

LIVING

PRODUCING

Waste water discharging

Garbage littering

Farming (Livestock farming)

Farming (Sugarcane burning)

Farming (Fertilizing & Pesticiding)

Fishing (Cage Culture)

HEAVY METAL POLLUTION — Cu, Pb, Cr, Cd

MICROPLASTICS POLLUTION — Polyethylene, Polypropylene

ORGANIC DEPOSITION — Feed Residues, Fish feces, Biocides

SECTION OF LAKE VICTORIA SHORELINE

The hydrological problems of Lake Victoria mainly result from the living and producing **behaviors of human beings.**



Credit: Wikipedia — INVASION



Credit: National Geography — POLLUTION



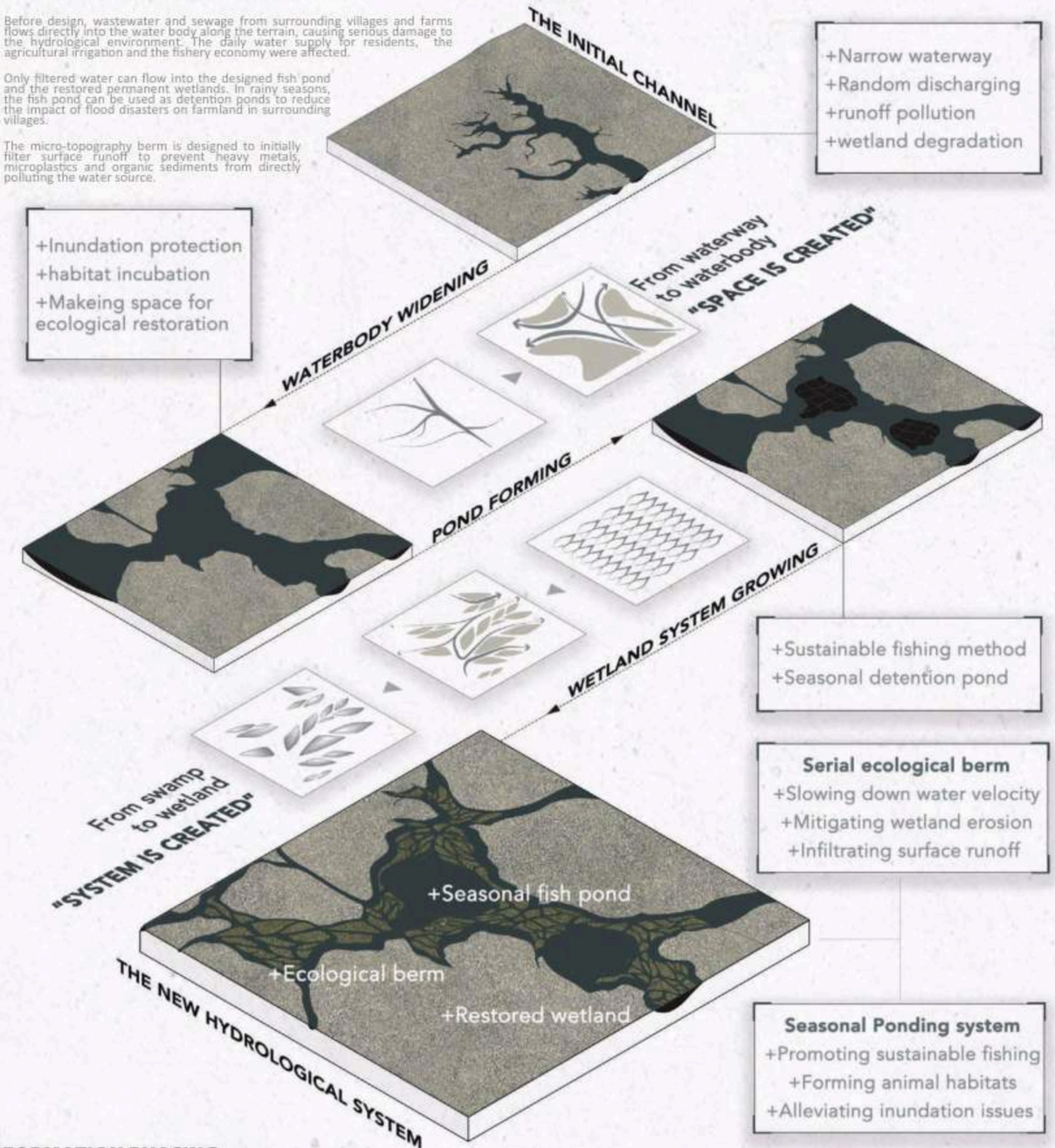
Credit: New Scientist — FISHERY

INTERVENTION DEDUCTION

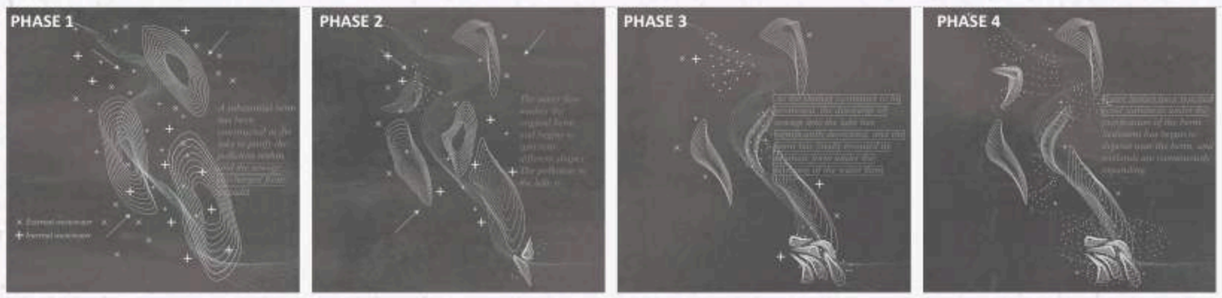
Before design, wastewater and sewage from surrounding villages and farms flows directly into the water body along the terrain, causing serious damage to the hydrological environment. The daily water supply for residents, the agricultural irrigation and the fishery economy were affected.

Only filtered water can flow into the designed fish pond and the restored permanent wetlands. In rainy seasons, the fish pond can be used as detention ponds to reduce the impact of flood disasters on farmland in surrounding villages.

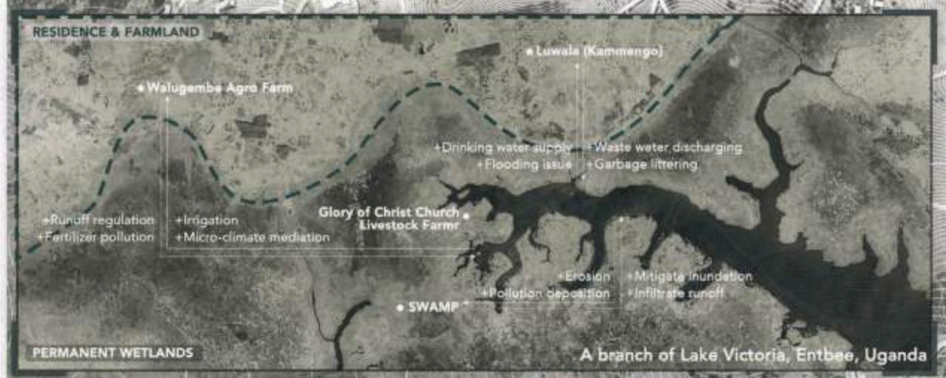
The micro-topography berm is designed to initially filter surface runoff to prevent heavy metals, microplastics and organic sediments from directly polluting the water source.



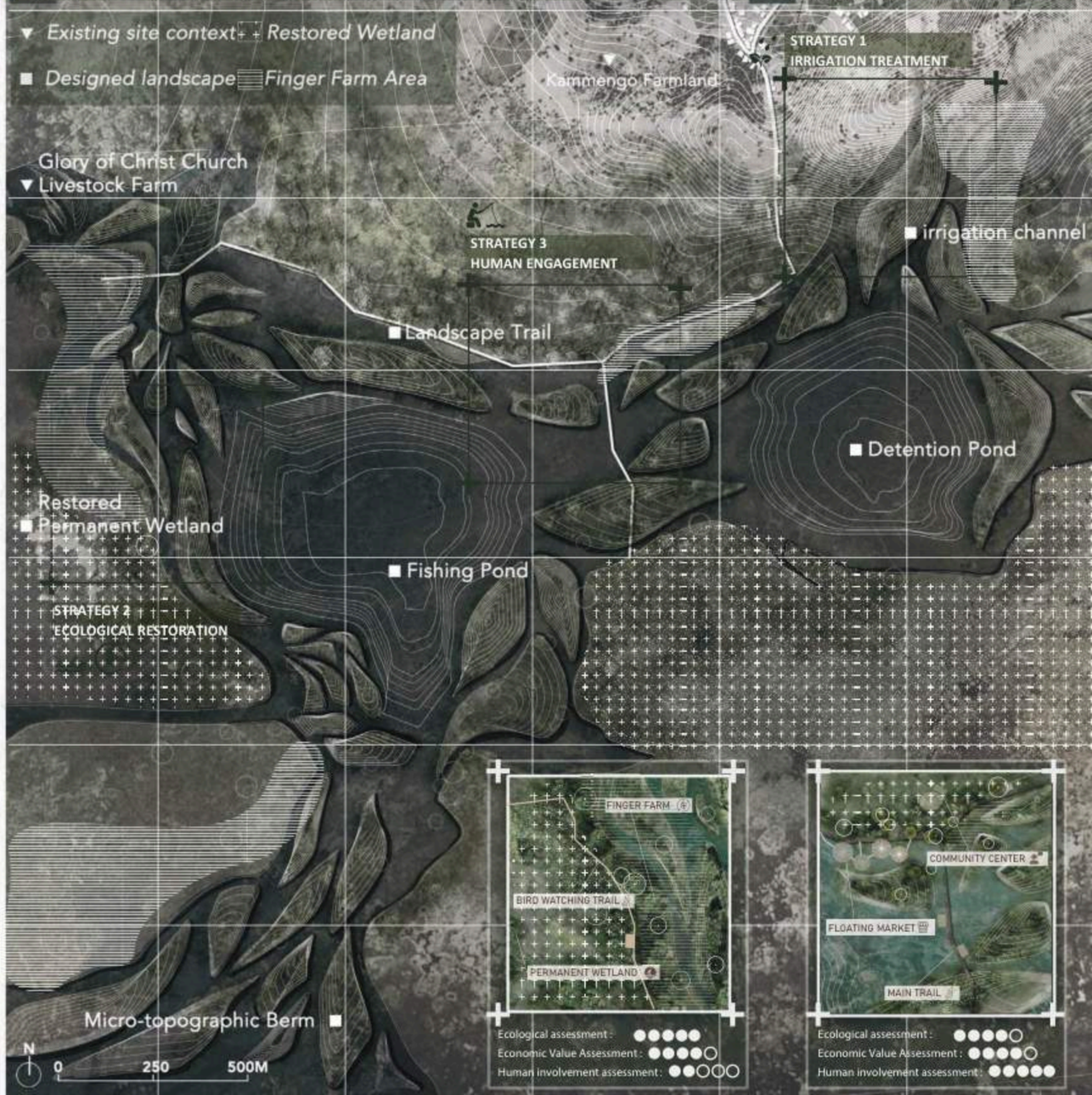
FORMATION PHASING



PROPOSED MASTER PLAN

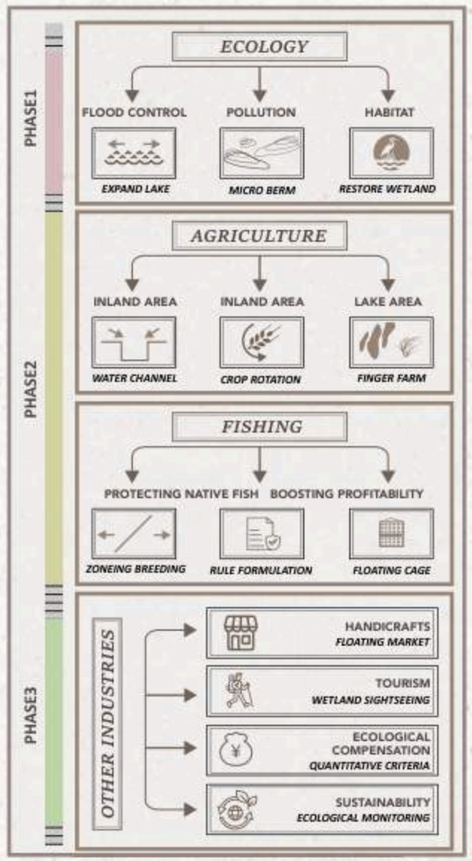




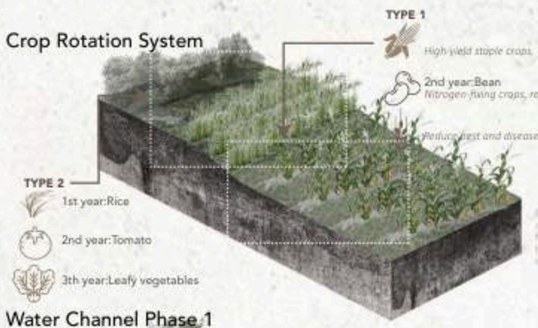


STRATEGY

Overall Strategy

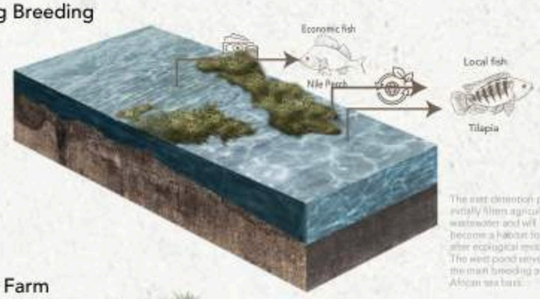


Crop Rotation System



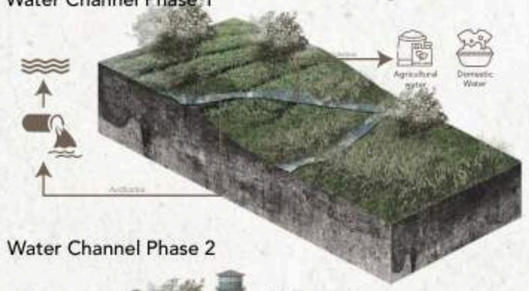
By scientifically combining crops, the use of chemical fertilizers and pesticides can be significantly reduced while optimizing soil fertility.

Zoning Breeding



The east detention pond initially filters agricultural wastewater and will later become a habitat for tilapia after ecological restoration. The west pond serves as the main breeding area for African sea bass.

Water Channel Phase 1



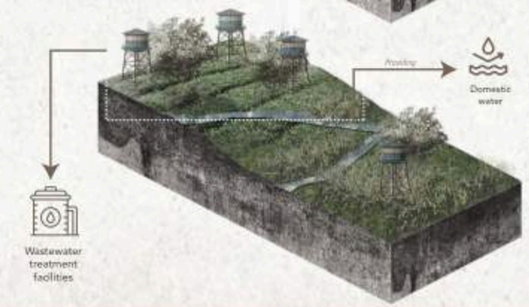
New water channels have been introduced into the farmland, initially used to collect agricultural wastewater and conduct preliminary purification.

Finger Farm



At the intersection of the wetland's edge and the main berm, a small-scale integrated fish and vegetable production system, known as the "finger pond," is established to enhance productivity.

Water Channel Phase 2



Once wastewater treatment facilities and lake water quality improve, these channels will serve as irrigation and domestic waterways, offering convenience to residents within the area.

Tourism Development

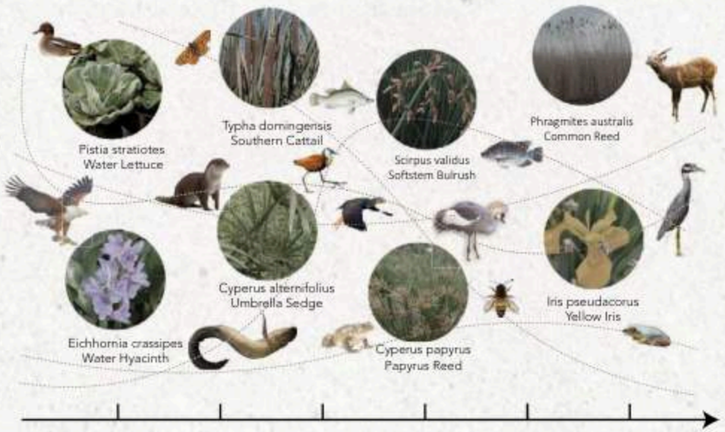


Developing tourism in the Victoria Lake wetlands diversifies the local industry, offering activities like fishing and wetland exploration.

ECOLOGICAL RESTORATION

Wetland Plant Selection

Plant native wetland species like papyrus, cattails, and reeds to filter pollutants from water bodies; construct eco-floating islands to offer new habitats for birds and fish; restore degraded wetland vegetation, allowing the ecosystem to gradually regain balance, thereby improving water quality.





Floating Island

Establish ecological floating islands to expedite wetland restoration and create additional habitats.



ECOLOGICAL COMPENSATION



The government provides various compensations to local residents to encourage them to switch to more environmentally friendly and efficient production methods while protecting wetlands.

FISHING	FISHING	AGRICULTURE	COMMUNITY
Ban Period Subsidy	Cage Support	Farmland Protection	Set Up A Fund
100 $ per month per household	500 fry per household.	$30 to $50 per hectare	Allocate 10%-15% to community funds
FISHING	**AGRICULTURE**	**AGRICULTURE**	**COMMUNITY**
Fish Fry Distribution	Reduce Fertilizer Use	Support New System	Job Support
500 fry per household	$50 for every ton of fertilizer reduced	$100 per hectare	Offer skill training and long-term job placements.

FLOATING MARKET



Set up floating markets using eco-friendly materials like palm leaves around the lake, designate specific open days to attract tourists to buy handicrafts and farm produce.



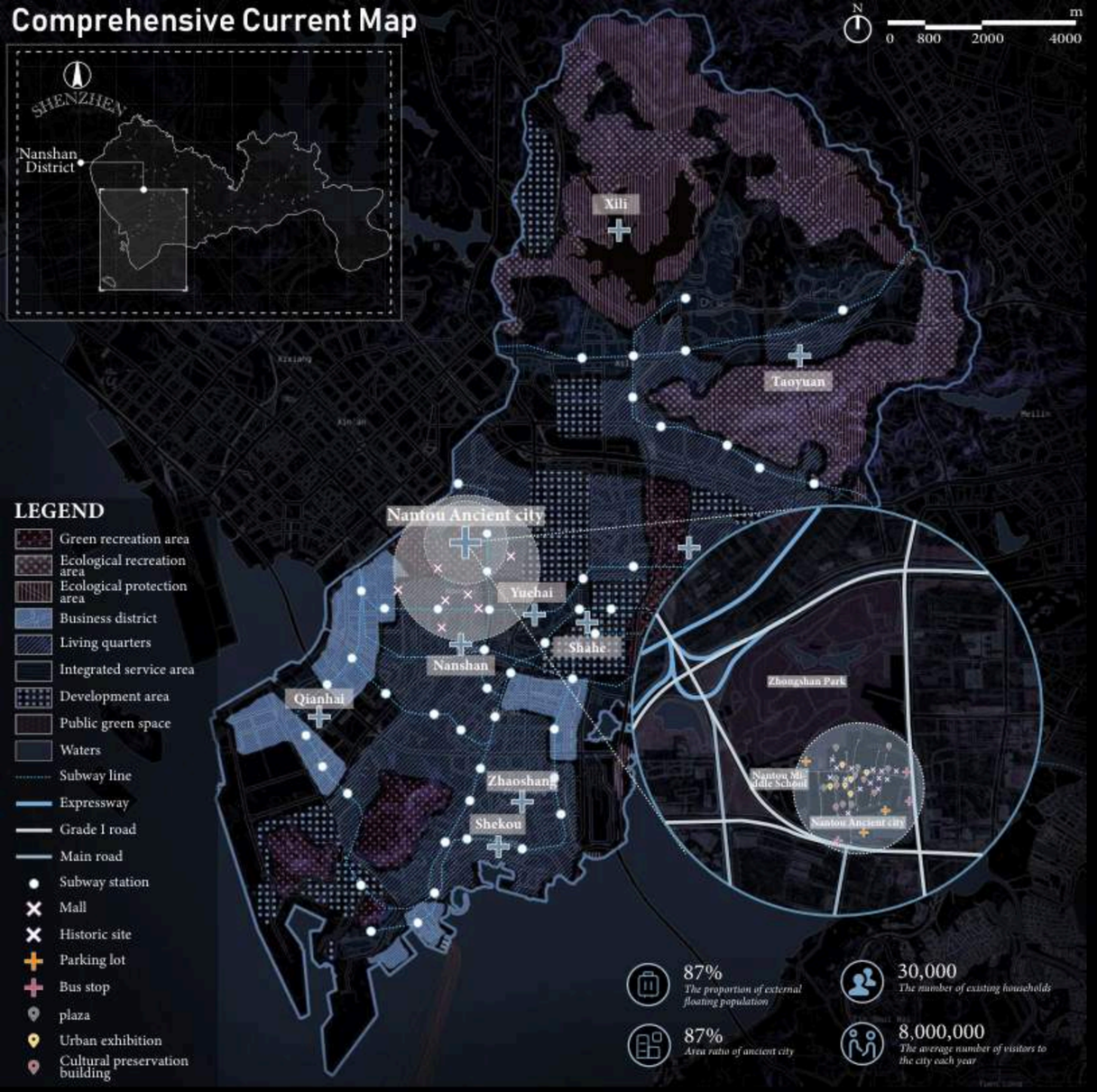
Neighbours But No Boundaries

Reimagining Future Living in Dense Urban Villages

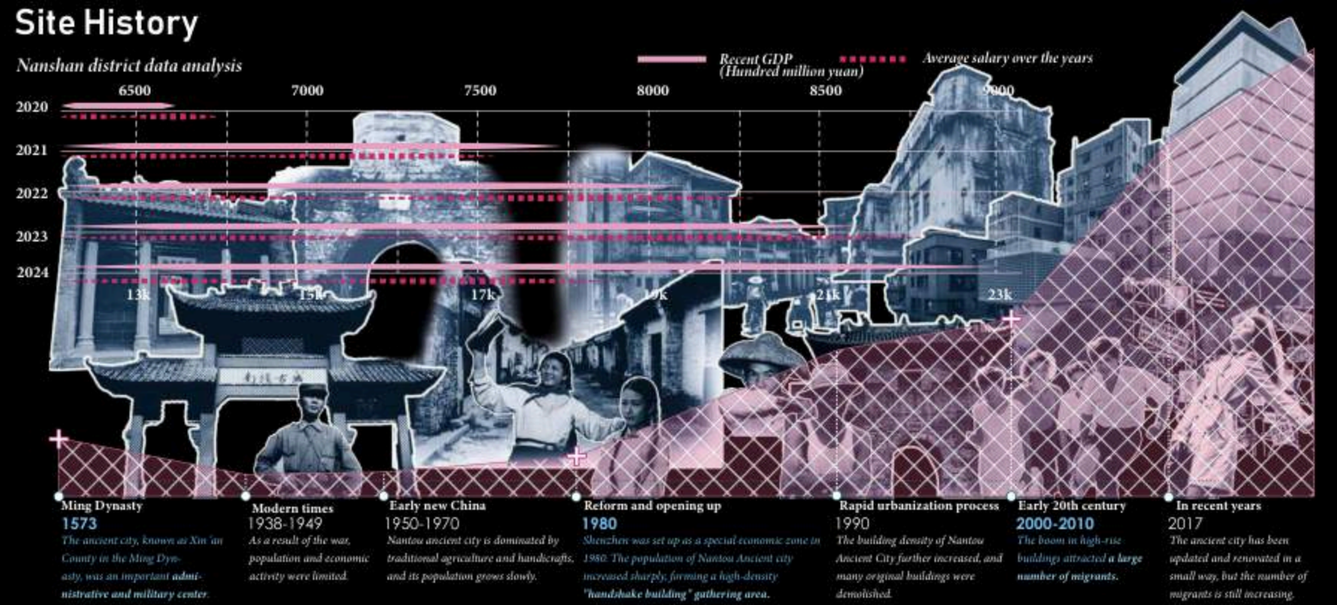
Landscape Architecture Design
Location: Shenzhen,China
Duration: 2024.07
Individual Work
Instructor: Kayden Xu

This project explores future urban living through the renewal of Nantou Ancient City in Shenzhen, balancing historical preservation with modern needs. It aims to create a high-density, functional, and livable environment, improve infrastructure, and foster shared public spaces, offering insights for similar high-density urban villages.

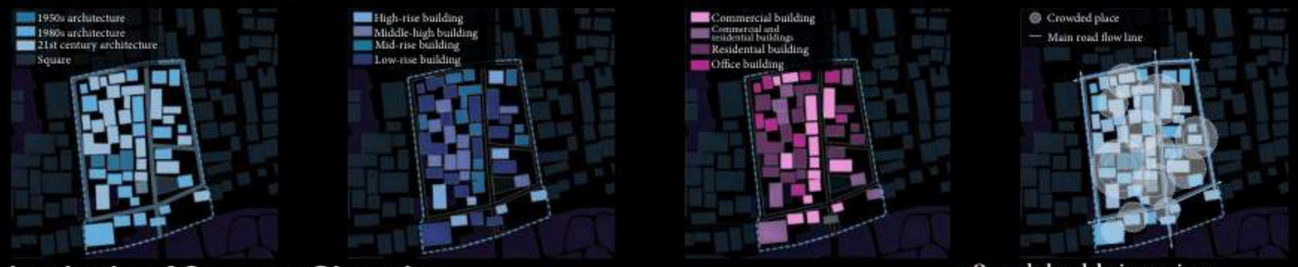
Comprehensive Current Map



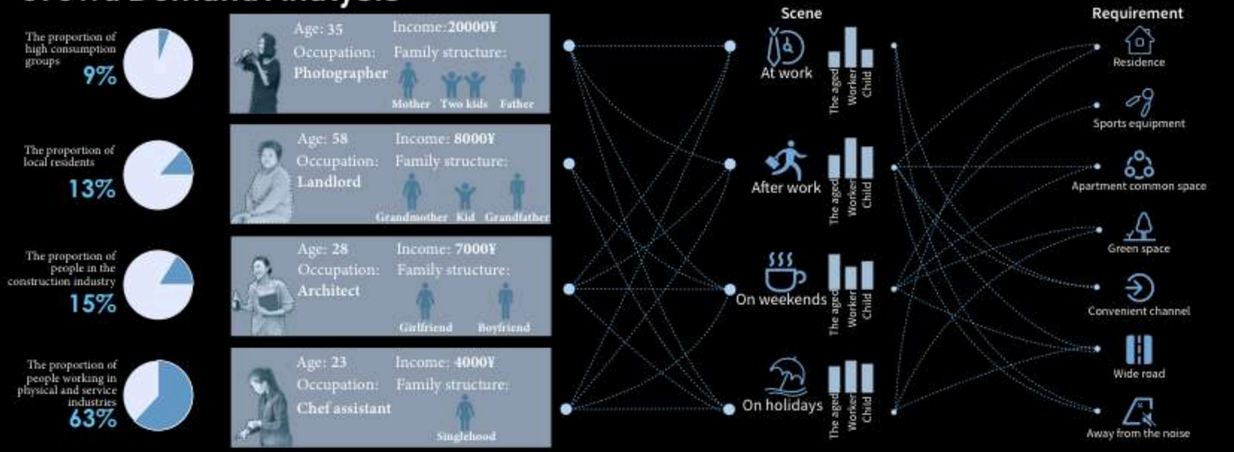
Site History



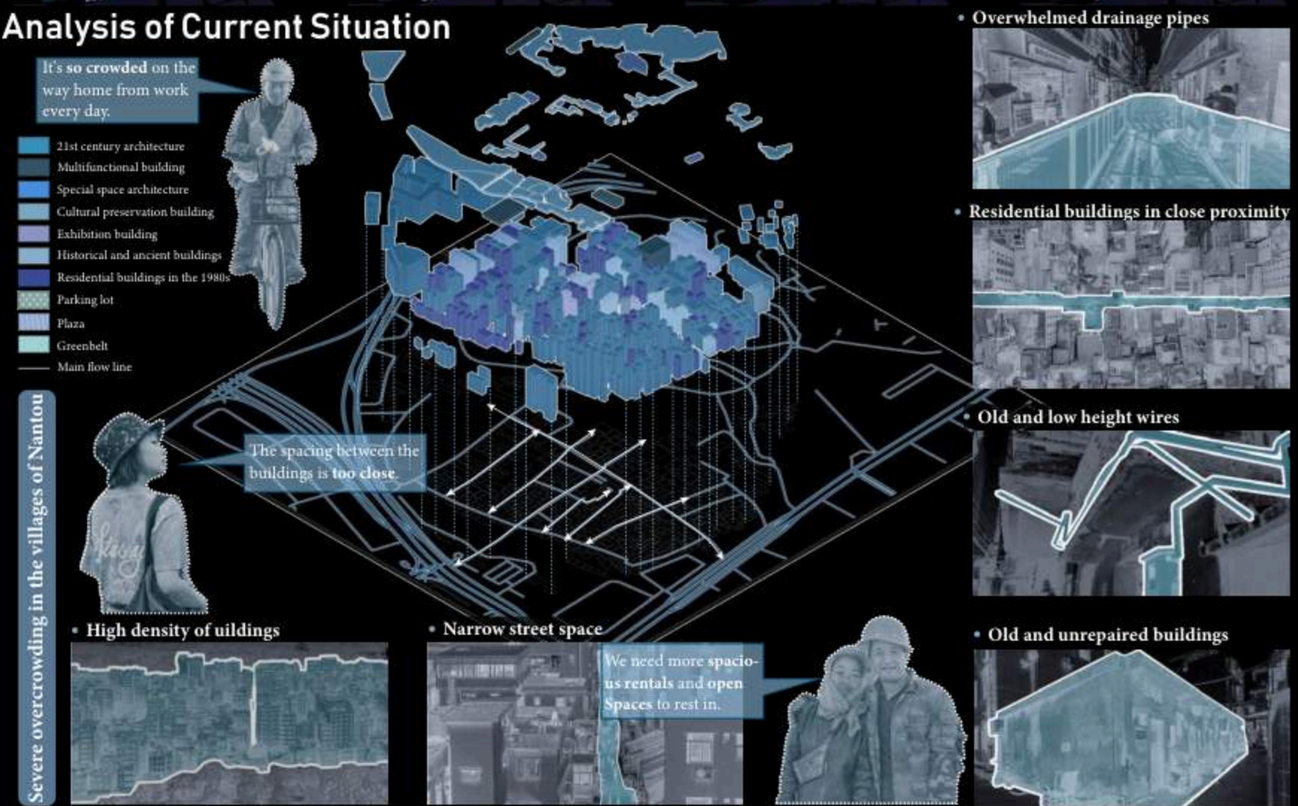
Site Foundation Analysis



Crowd Demand Analysis



Analysis of Current Situation



Overall Strategy

Neighbors have no boundaries

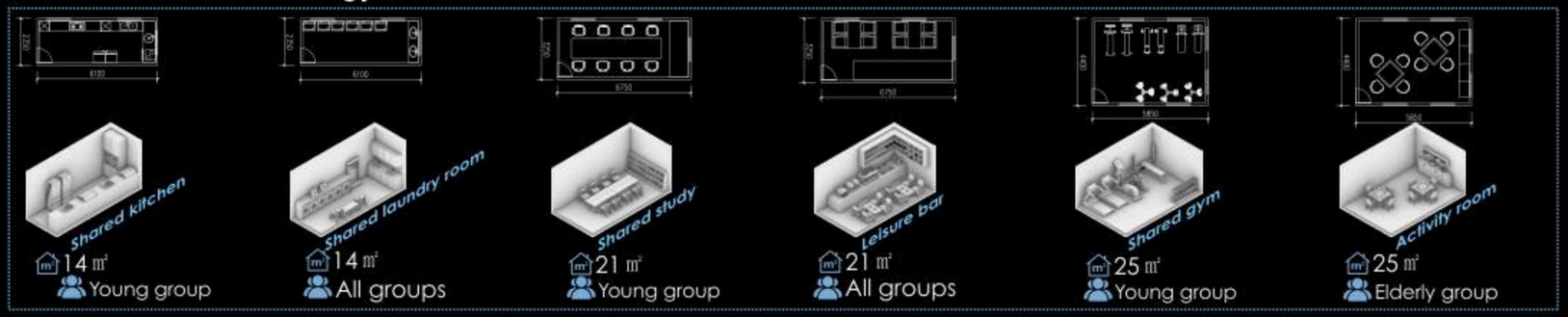
有邻
Single house living mode

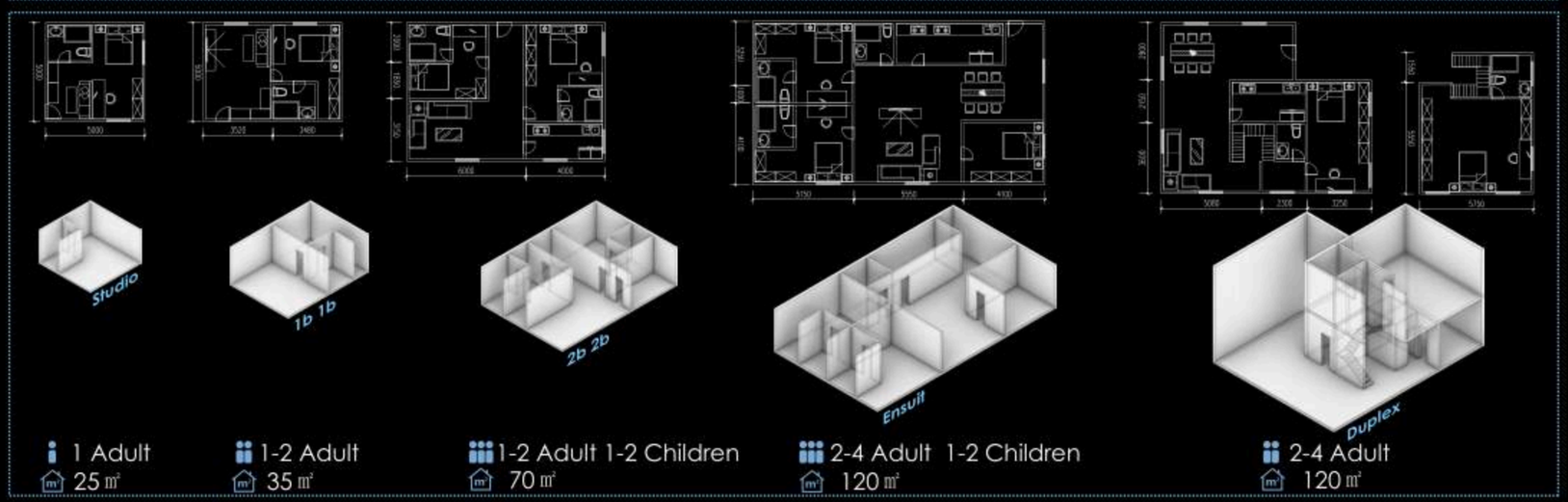
无界
Overturn the shared model of borders

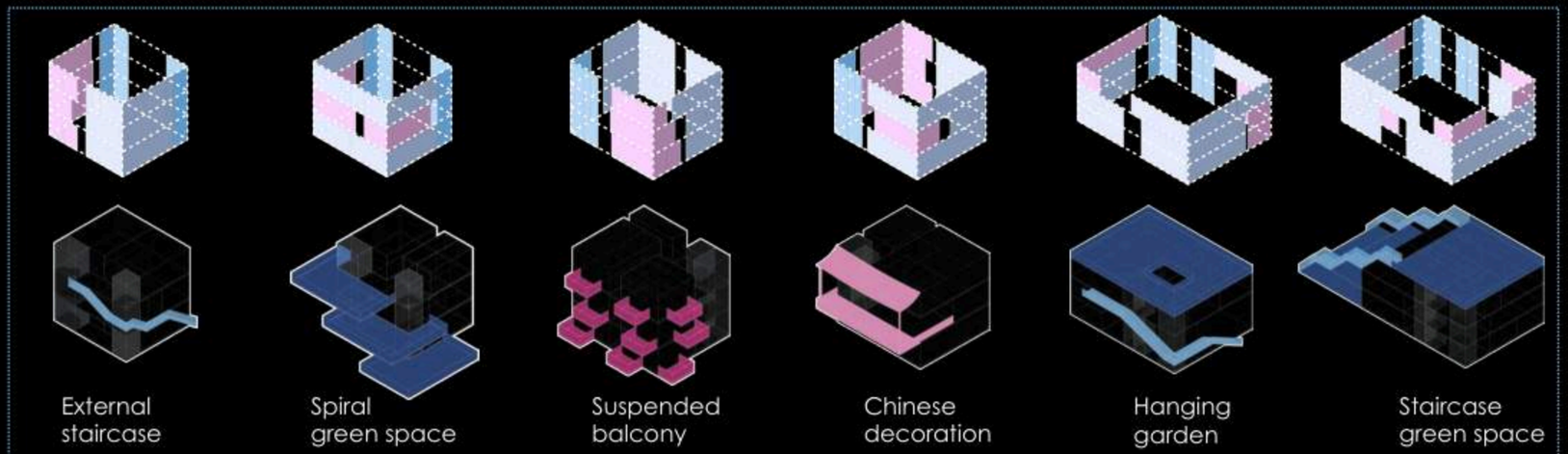
- Private residence
- Open space
- Shared module
- Road connection

- **a** *Building organic group*: ***sharing patterns*** of different scales
- **b** *Activating the ground floor space*: accelerating ***road connectivity***
- **c** *Extraction of regional cultural elements*: patterns and architectural components extracted from ***Chinese gardens***
- **d** *Horizontal + vertical co-housing of young and old*: a new three-dimensional ***spatial living model***
- **e** *Shared catalysts*: Sky courtyards, terraces and roof gardens create ***new green Spaces***

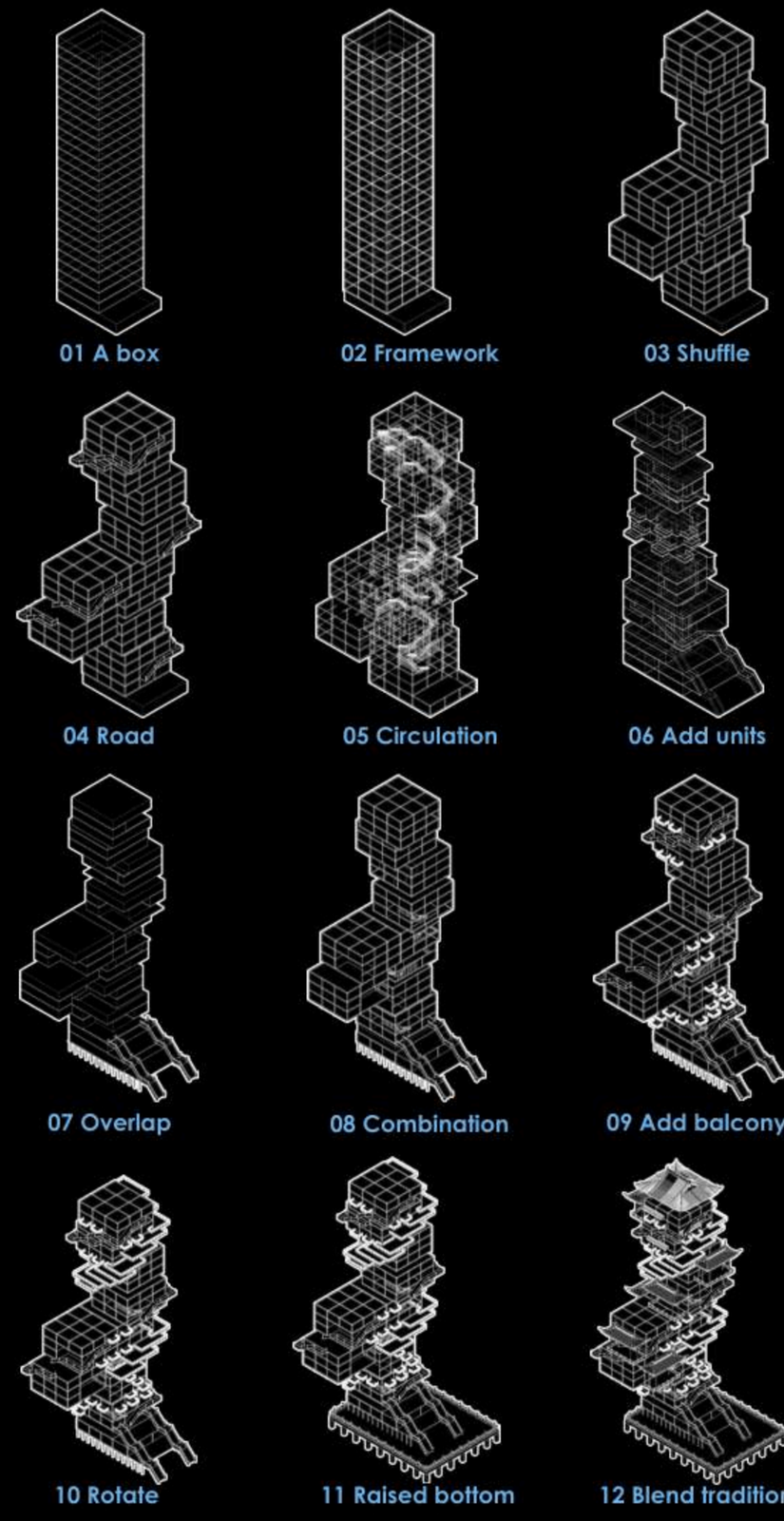
Reconstruction Strategy



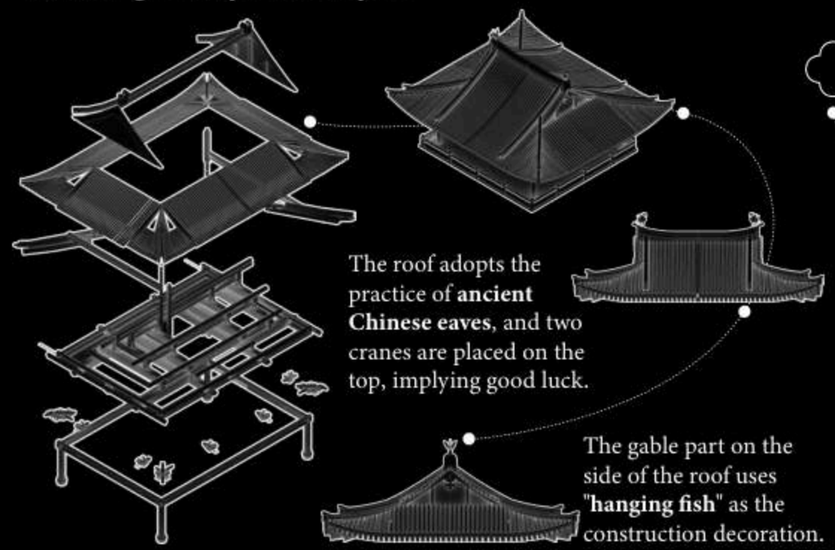




Generative Process



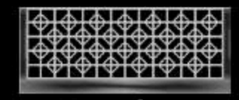
Building Sample Analysis



The roof adopts the practice of **ancient Chinese eaves**, and two cranes are placed on the top, implying good luck.

The gable part on the side of the roof uses "**hanging fish**" as the construction decoration.

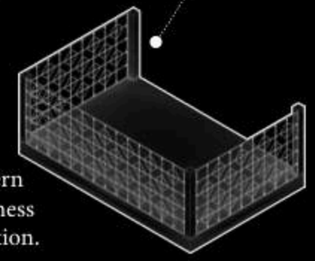
Chinese traditional pattern: **begonia pattern**



Evolved into a guardrail style



Chinese traditional garden window style: crabapple flower window



This decorative pattern is a symbol of happiness and extreme satisfaction.

Section



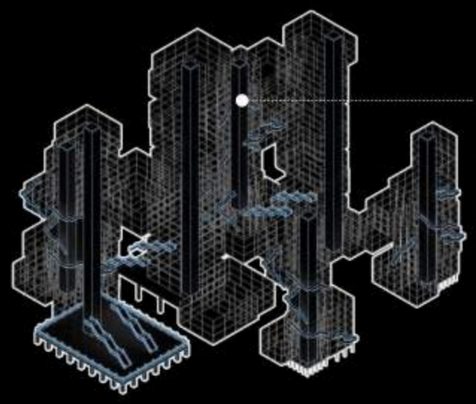
Illumination analysis

Building lighting directly benefits from large Windows.

Before

After

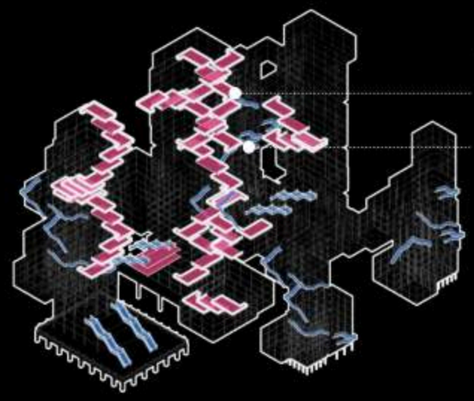
Building Program Analysis



Vertical internal core

Core barrel structure

The basic part of the interior of the building is the **vertical elevator and stairs.**

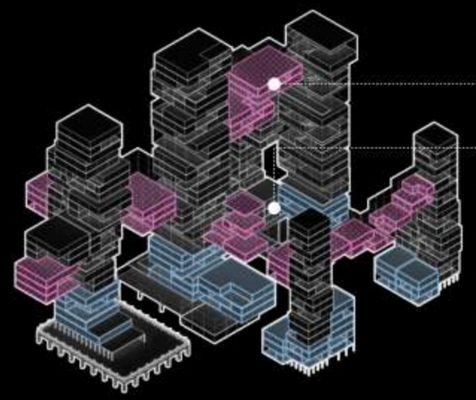


Outer cycle space

Spiral green space

Exterior stair

By adding structure to the exterior of the building, the **green space and access** are reorganized to form a new circulation.

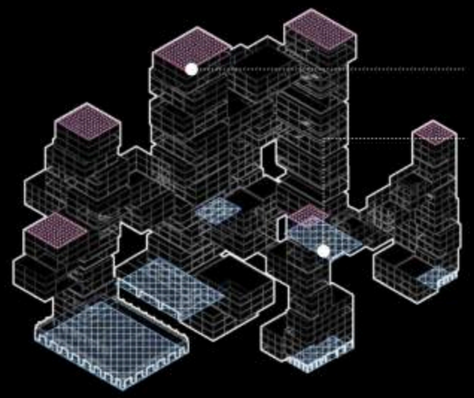


Public space

Shared function area

Commercial area

The public space is arranged according to the crowd flow line, with **the commercial space at the bottom.**

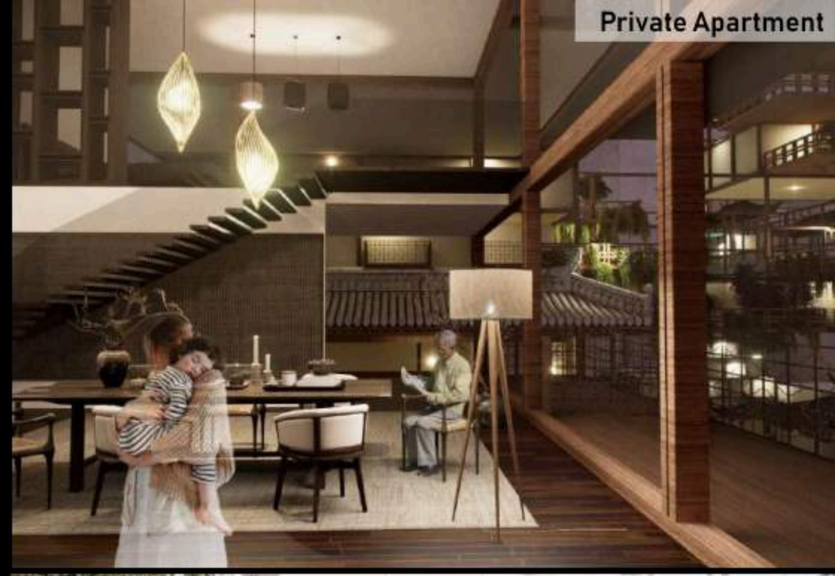


Open floor space

Space on the roof

Bottom shelf

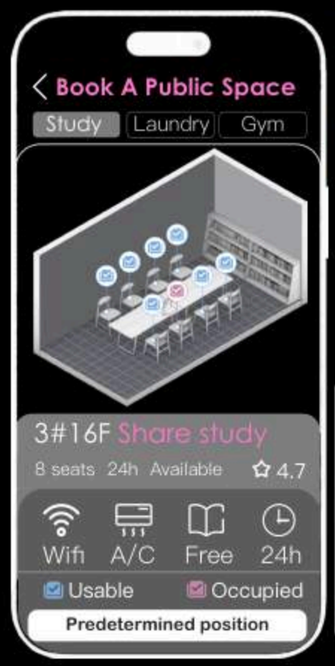
The **base of the building** was **raised** so that the narrow street was widened for easy passage.





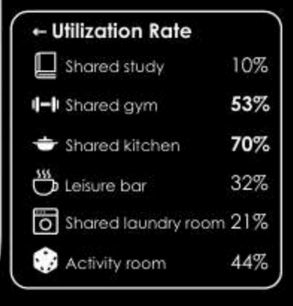


Interactive analysis



Interaction

Residents can **reserve and use the shared space on the app** according to their needs.





Activities

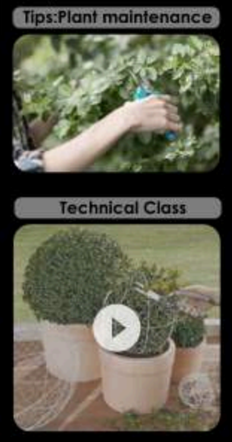
Community art events can be viewed and attended on the app.





Information

Information and conservation advice for small farms on the app.



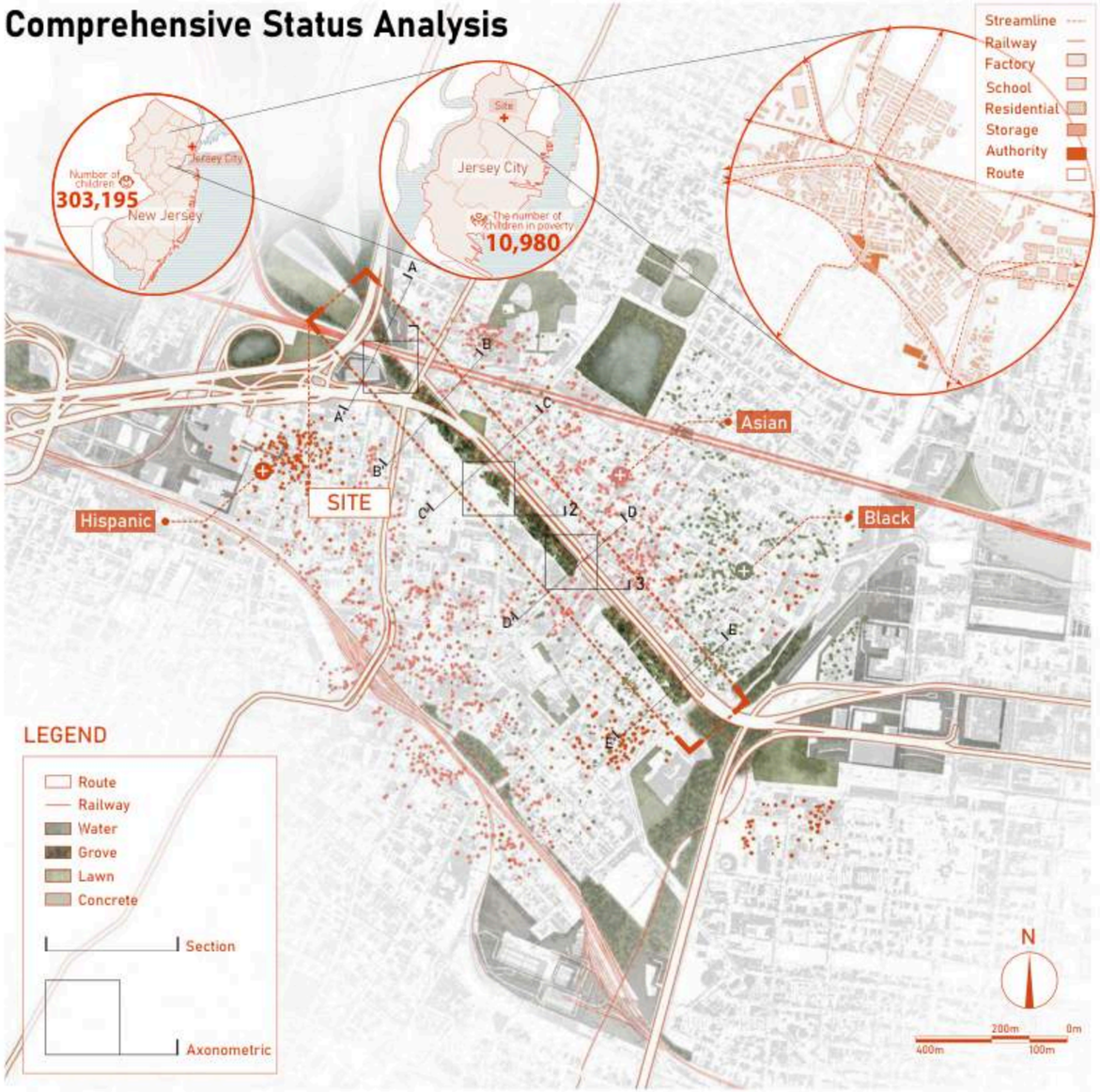
PlayCube

Multicultural Children's Landscape Park

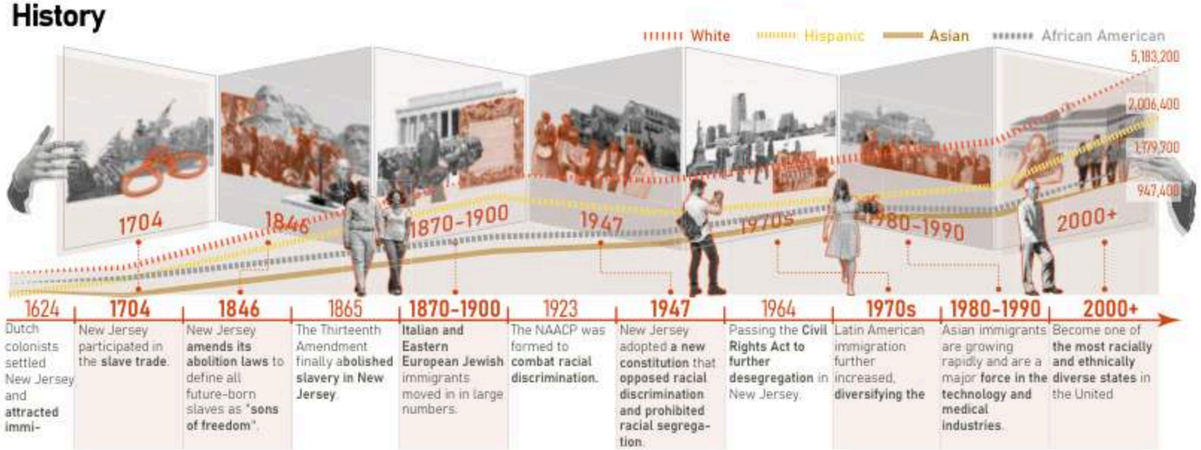
Landscape Architecture Design
Location:New Jersey,USA
Duration:2024.07
Individual Work
Instructor:Jennifer Birkeland

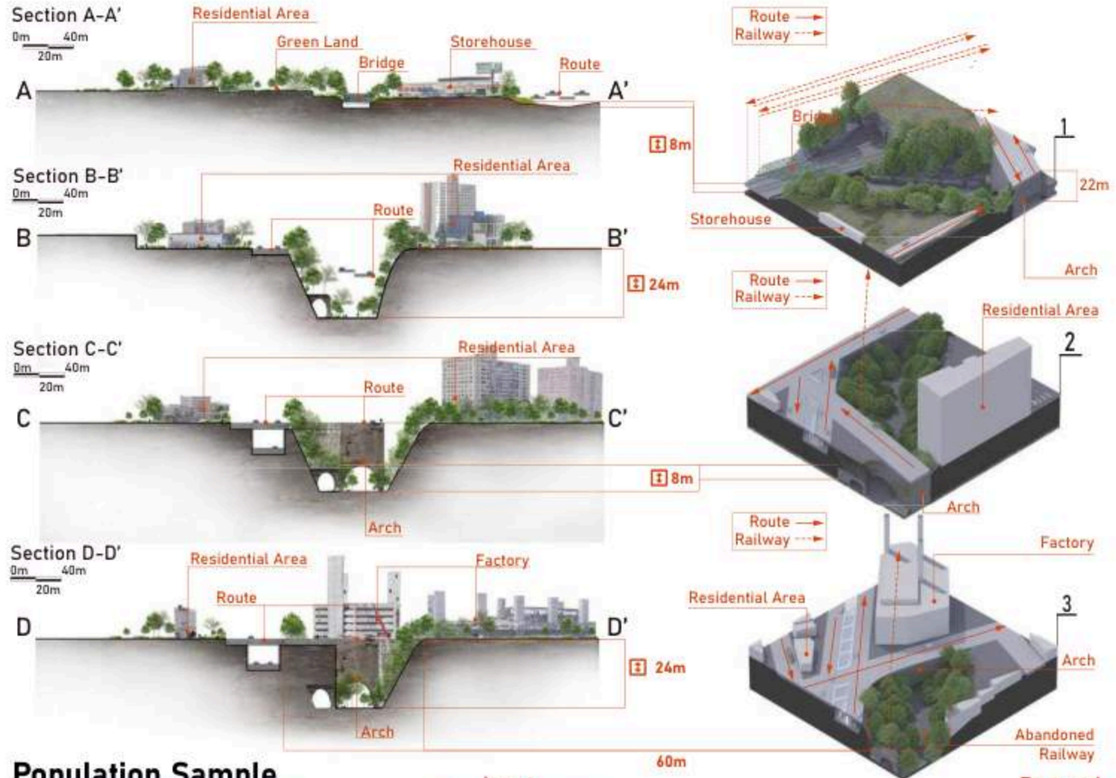
PlayCube is a multi-dimensional park featuring elevated pathways, climbing nets, and slides that connect children of diverse cultural backgrounds. Designed to foster friendship and unity, its inclusive play spaces encourage collaboration and shared experiences, while cultural motifs and vibrant colors celebrate diversity and build bridges of understanding.

Comprehensive Status Analysis



History





Population Sample



Liam
Race: African American
Age: 2
Gender: Male

Juliana
Race: Hispanic
Age: 7
Gender: Female

Adam
Race: White
Age: 10
Gender: Male

Ansuya
Race: Asian
Age: 14
Gender: Female



Issue

Social Disorder

Loneliness

Racial Bullying

Cognitive Delay

Developmental Delay

Mental Illness

Reason

Guardian neglect

lack of green space

Ethnic complexity

Inequality

Demand

Integration

Social

Cognition

Sports

Nature

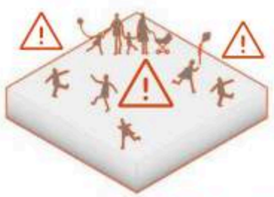
Existing Problem



1 **Very high height difference**, the site is difficult to use.

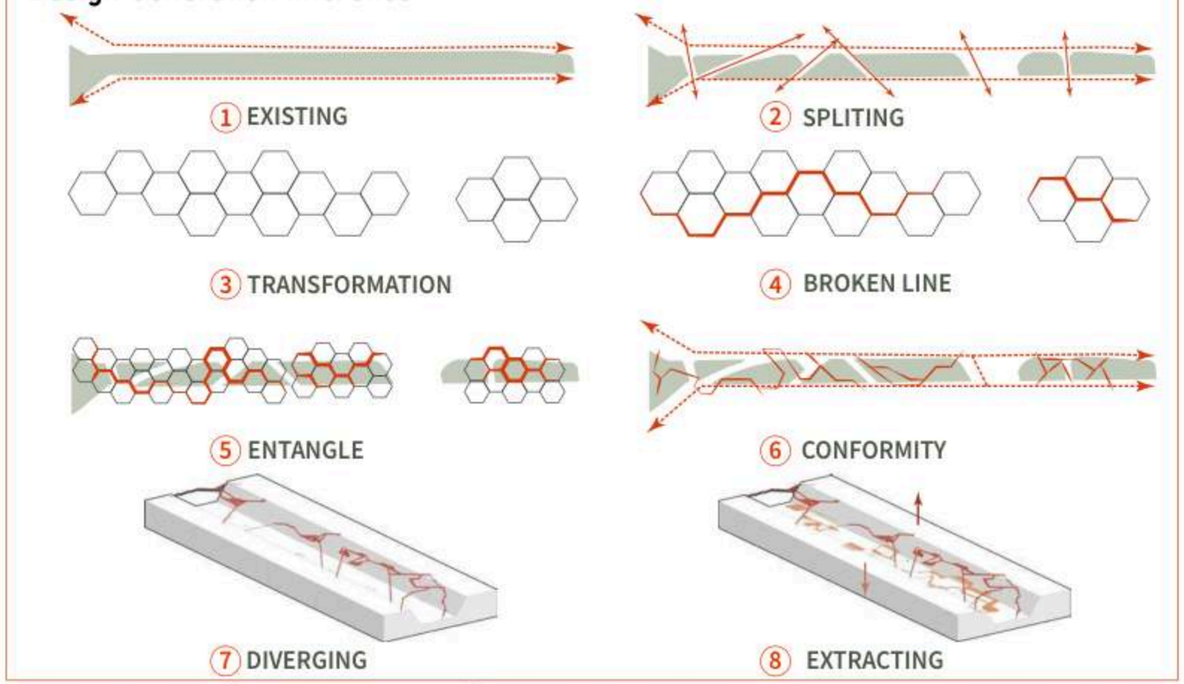


2 **Racial segregation**: Lack of human connections



3 High risk factor, **lack of children's public**

Design Generation Inference





Children of different ages have different cognitive development level,

Jean Piaget
Swiss psychologist

"Children's paradise for all ages"

User : Children

Not yet racially prejudiced

1 3
2 4

Promote Communication

Build Relations

Demand for public recreational facilities

Different Cognitive Development Stages

Site

Site Characteristics: **Urban Abandoned Railway**

Sunken — Isolation — Cell Grid — Space Exclusivity

Hierarchical Unitized Landscape Space

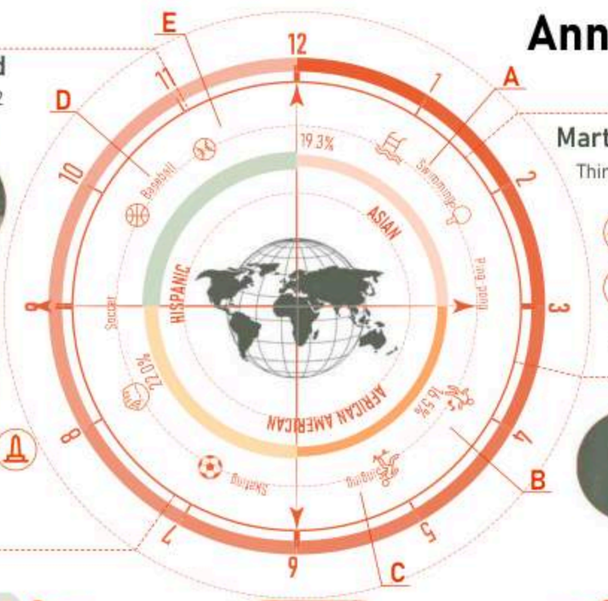
Cognitive Development

Break Racial Segregation

1 Sensorimotor Stage	2 Preoperational Stage	3 Concrete Operational Stage	4 Formal Operational Stage
0 -2	2-7	7-11	12+
Sensory Experience	Abstract Thinking	Concrete Thinking	Reasoning Thinking
Racial Identification	Racial Interaction	Racial Relations Establishment	Racial Integration

Annual Festival

Celebration Season



The Day of the Dead

November 1 and 2

Reunion

Belong



Origin Nation: Mexico

Origin Nation: America



Values

Comm-emorate

Fourth of July

Independence Day

Martin Luther King Jr. Day

Third Monday in January

Liberty

Equality



Origin Nation: South Africa

Holi

The Middle of March



Diversity

Celebration

Origin Nation: India

Issue

- Racial Misunderstanding
- Isolation
- Cognitive Impairment
- Developmental Delay

Solution

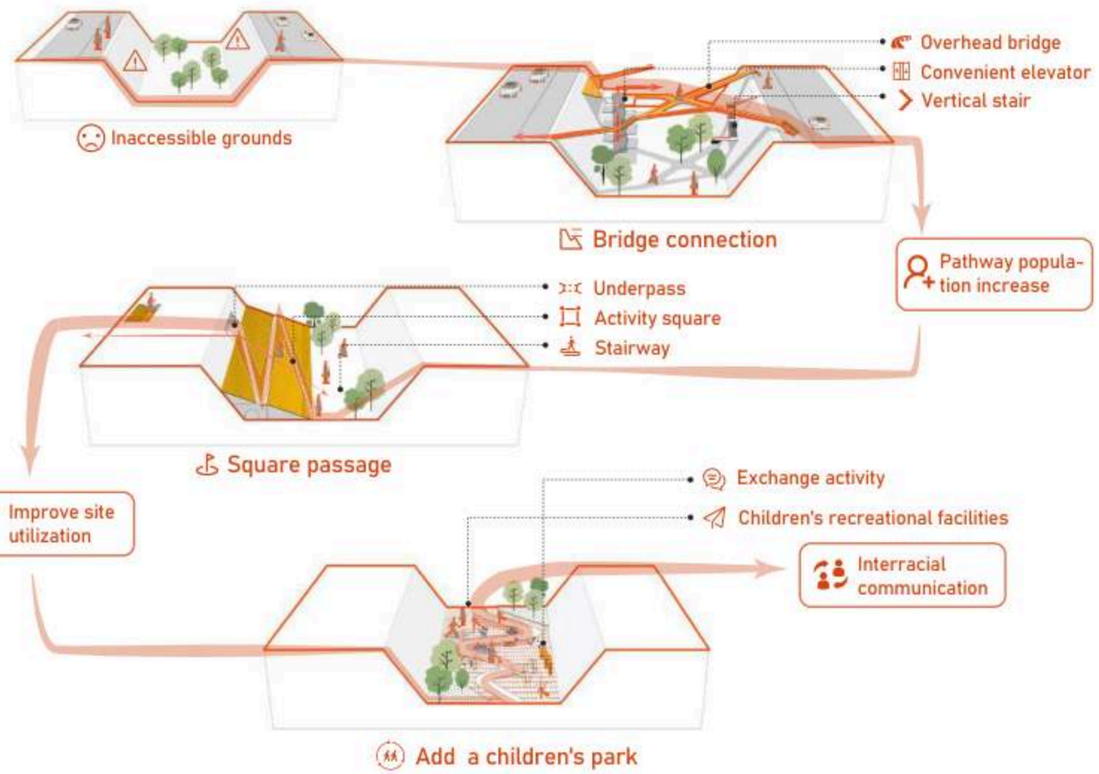
1 2 3 4
A B C D E

Festival

Goal

- Integration
- Social
- Cognition Development
- Physical Development

Design Strategy



Physical Model



Size: 900mm*600mm
Material: Basswood board , filament
Scale: 1:600

N

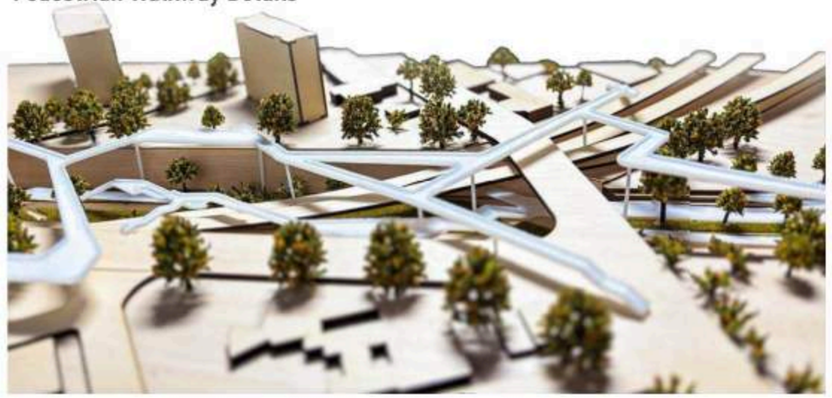
The Pedestrian Bridge was made using 3D printing, with basswood used as the main material for the rest of the structure.

This handmade model includes a friendship splash zone and adventure play plaza as well as three-dimensional bridge nodes.

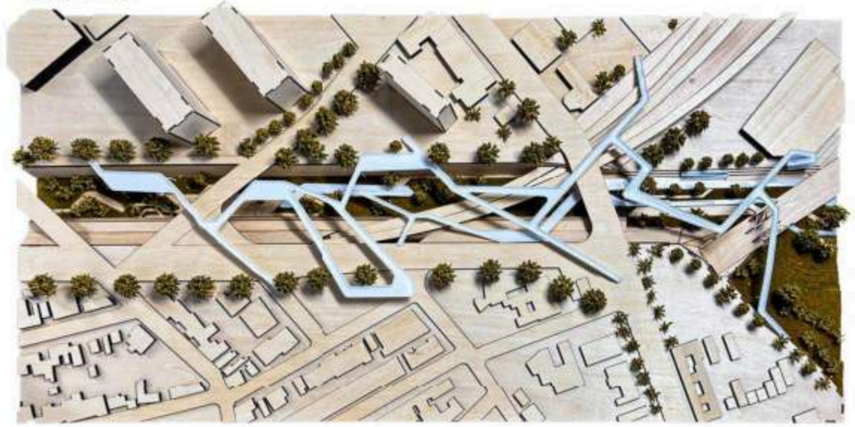
Overall Model



Pedestrian Walkway Details

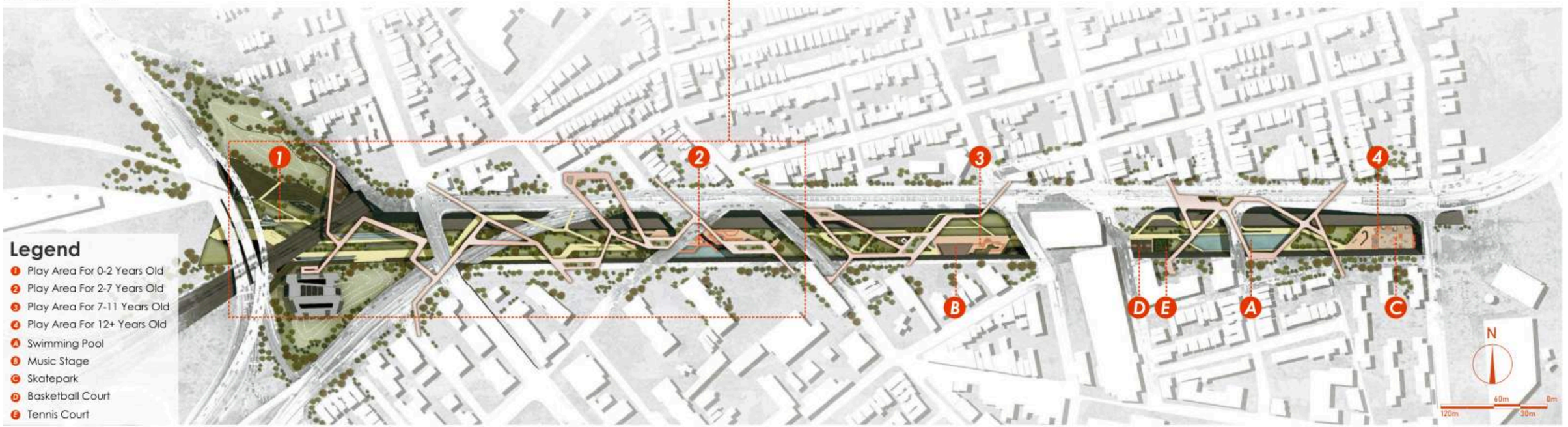


Aerial View



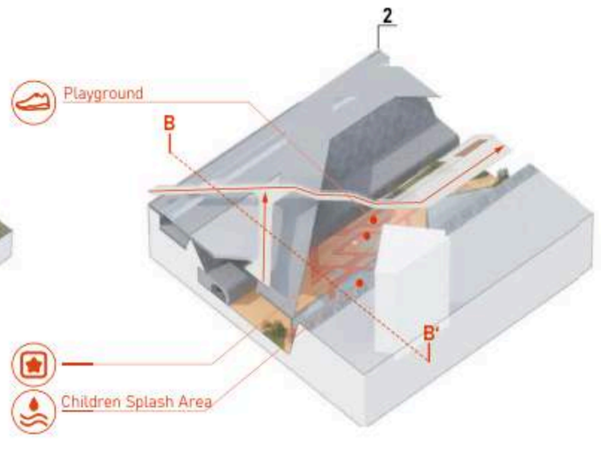
Scope of Model Making

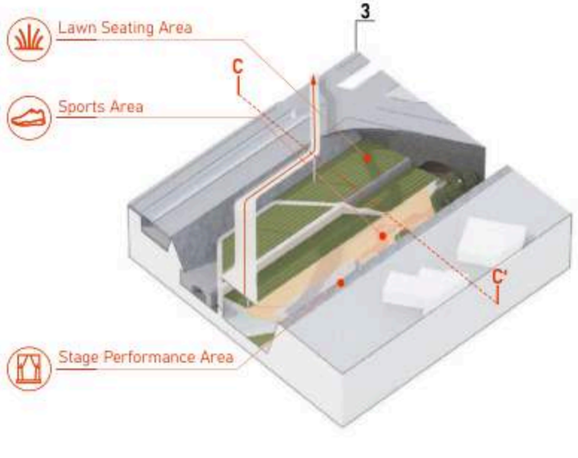
Master Plan



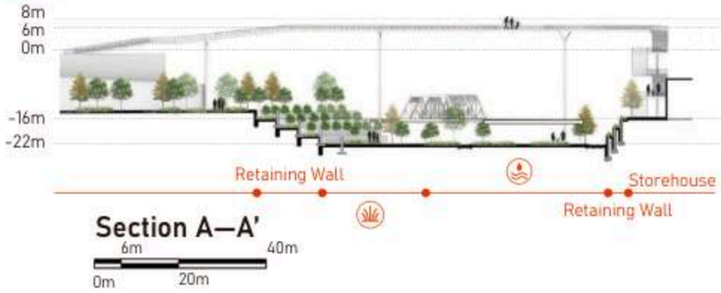
Sectional Design

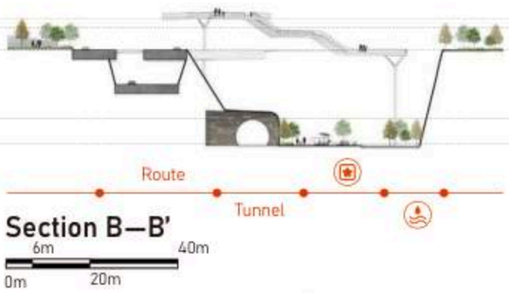


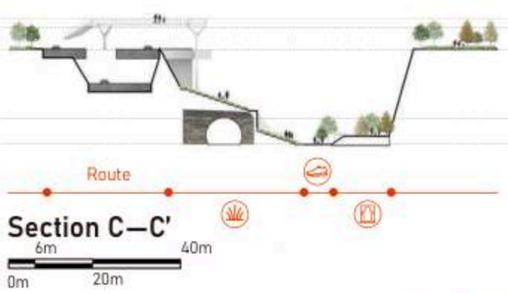


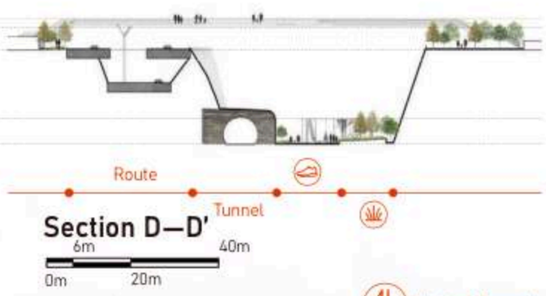








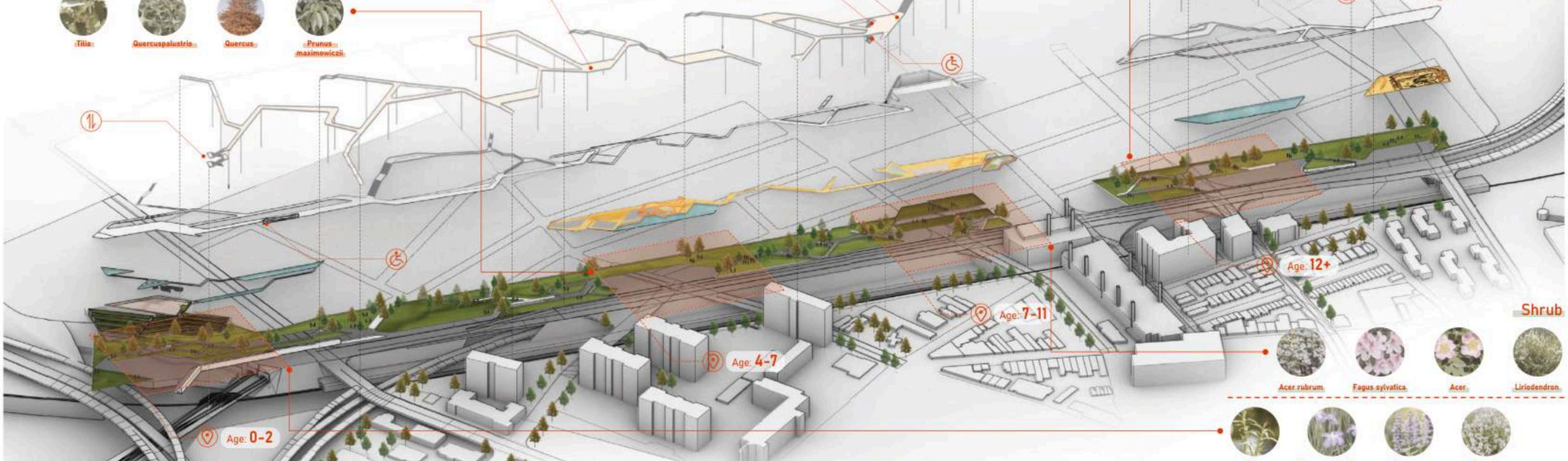




Macrophanerophytes

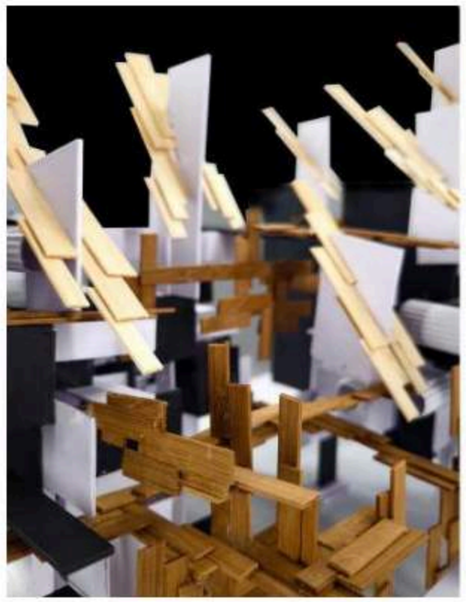


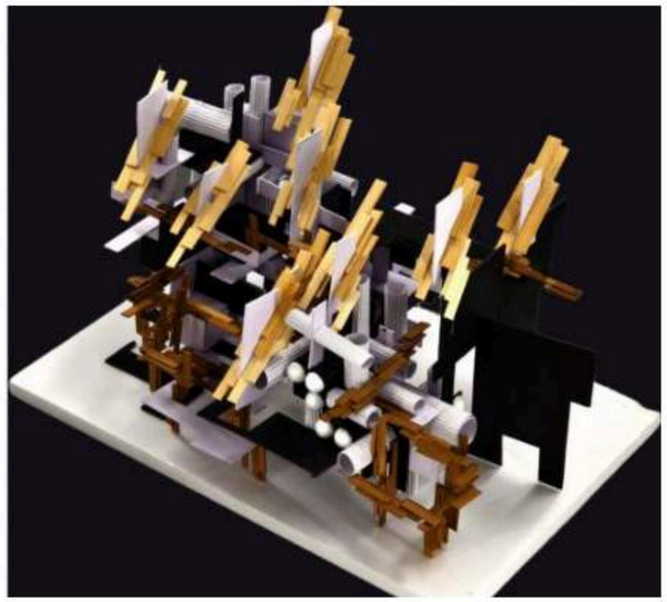




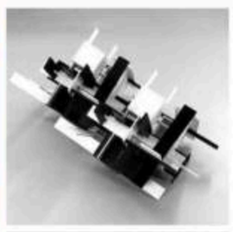
OTHER WORKS

1.REWEAVING

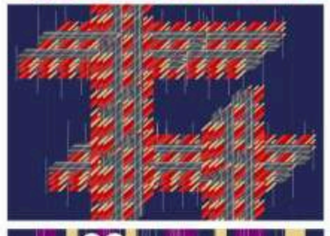


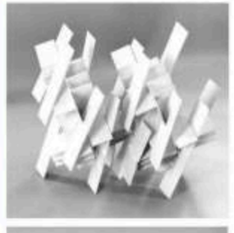


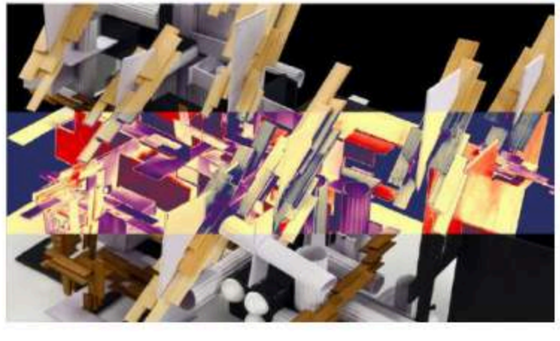


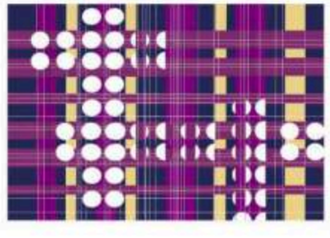














2.GAME DESIGN













Rest In Winter

Children's well-being during war

Let the children cultivate their observation in the interaction with Anna, and enjoy the most warm parent-child time.In-discriminate expression of anti war stance, so that children can understand the world while cultivating empathy.









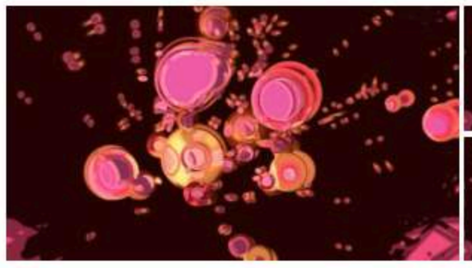


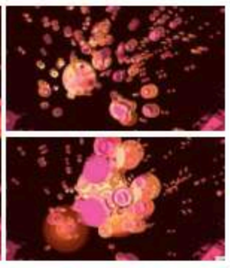
Mumbo Jumbo

Assist children with dysphonia

The game is designed with Unity and uses the role-playing role of the protagonist, a little girl, to understand the process of vocalization, become familiar with the principles of vocalization, and assist in vocalization. Treatment of children's aphasia through games

3.BUILDING FASADE INSTERACTIVE DESIGN







TouchDesigner Program

